

Dyadic Multiresolution and Product-Series Representations in the Fabius–Rvachev System

**Rational mass arrays, Haar–Schauder expansions, Walsh–Thue–Morse products,
beta-mixture limits, Bell–Bernoulli scale energies, and inverse-quantile duality**

A repository-aware frontier report for Vladimir Reshetnikov’s ProveIt project

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Abstract

The existing ProveIt corpus develops the Fabius function, its inverse, Rvachev’s compactly supported up-function, the Thue–Morse sequence, exact dyadic values, q -binomial formulas, infinite sinc products, Bell and Bernoulli polynomial machinery, and phase-resolved Lambert– W endpoint asymptotics in considerable depth. This report first organizes those results into a representation atlas and then develops a complementary multiresolution layer that was not located in the audited LaTeX corpus.

Let $p = F'$ and let $\Delta_{N,k} = F((k+1)2^{-N}) - F(k2^{-N})$ be the exact dyadic cell masses. The same rational projective array produces four different finite models. It produces a fixed-cell uniform mixture for p and up, an exact rational unnormalized Haar expansion of p , a rational Faber–Schauder expansion of F , and the finite Walsh–Hadamard transform of p . Their Fourier images yield locally uniform limits of rational exponential polynomials multiplied by sinc windows. The Walsh transform of every basis function is itself a finite product of sines and cosines. Consequently the infinite sinc product admits a new series whose summands are finite trigonometric products and whose coefficients are exact Walsh transforms of dyadic Fabius masses.

A Walsh mode indexed by m has exactly $\text{wt}_2(m)$ vanishing algebraic moments. Writing $\sigma(m) = \sum_j j b_j$ for the positions of the nonzero binary digits, its first nonzero moment is

$$\int_0^1 x^{\text{wt}_2(m)} w_m(x) dx = (-1)^{\text{wt}_2(m)} \text{wt}_2(m)! 2^{-2 \text{wt}_2(m) - \sigma(m)}.$$

Summing over a fixed binary weight produces the Gaussian binomial coefficient $\begin{bmatrix} N \\ r \end{bmatrix}_{1/2}$. Higher Walsh moments are encoded by a finite Bell-polynomial expression whose logarithmic coefficients are explicit Bernoulli-number sums. This gives a direct bridge among Walsh cancellation, Thue–Morse signs, q -binomial coefficients, and the Bell–Bernoulli structure of the Fabius moments.

A second positive approximation is obtained by differentiating the Bernstein polynomial of F : p is the uniform limit of finite mixtures of beta densities with rational weights $\Delta_{N,k}$. Its moment-generating and Fourier transforms are therefore limits of positive rational mixtures of confluent hypergeometric functions. A dual construction uses dyadic values of the inverse $G = F^{-1}$: equal probability cells with variable spatial widths give a finite sinc-mixture formula for $\widehat{\text{up}}$, as well as polynomial limits for all moments.

Finally, the exact Haar detail energy E_N is rational and has an all-orders continuum expansion

$$E_N \sim \frac{h^2}{16} \|p'\|_2^2 - \frac{h^4}{384} \|p''\|_2^2 + \frac{h^6}{20480} \|p'''\|_2^2 - \cdots, \quad h = 2^{-N}.$$

The complete coefficient sequence is generated by sinc^4 , hence by complete Bell polynomials in Bernoulli numbers. The beyond-all-orders aliasing remainder provides a new place where the

known log-periodic Fourier decay should reappear. The report distinguishes repository input, results proved here, numerical checks, and conjectures; “new” always means new relative to the audited repository corpus, not an assertion of worldwide priority.

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1 Audit protocol, scope, and novelty boundary

1.1 Pinned corpus and cumulative audit

The documentation tree

<https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs>

was audited recursively. The Git tree contained 89 files with extension `.tex` at the source-level inventory used in the audit. During this investigation the branch advanced; the report is pinned to commit `924652f4a9a656fd957d2936b2f053747c6f6c47`. The five-commit delta from the preceding pin modified the primary exposition, its coverage ledger, and an integral-transform report, but added no new TeX path. The delta was checked before the novelty boundary below was frozen.

The audit is cumulative rather than a naive count of duplicated revisions. Earlier source-level passes covered the complete path inventory; the present pass reread the canonical primary exposition, current audit ledgers, active frontier syntheses, and every recent report capable of overlapping the proposed results. Historical copies under `archive/` were normalized against their live successors. Imported papers were used as literature baselines, not counted as independent repository discoveries.

What the current formal source already contains

The current primary exposition and Lean modules include the probability law of weighted uniform series, finite atomic and spline approximants, exact dyadic values, quantile transport, sinc and cosine products, Fourier inversion and Poisson summation, moment recurrences, a Fourier–Legendre expansion, normalized integer Volterra calculus, inverse-Fabius infrastructure, and the proved Lambert-phase endpoint machinery. These are inputs below. Repackaging one of those statements with different notation is not counted as a new result.

1.2 Explicit exclusion map

The following recent manuscripts were treated as an exclusion map, not as a source of claims to relabel.

Already developed in the corpus	Excluded from novelty claims here
Primary Fabius–Rvachev synthesis	Function triad, probability construction, refinement laws, exact dyadic arithmetic, sinc product, Fourier inversion, moments, Fourier–Legendre series, and formal Lambert asymptotics.
Integral and transform reports	Weighted antiderivatives, inverse-pair area identities, Mellin/Newton continuation, negative moments, fractional primitives, order statistics, Stieltjes/Cauchy transforms, autocorrelation, and Gini-type integrals.
Spectral determinant and Barnes reports	Zero multiplicities, spectral zeta, Fredholm determinants, heat traces, Barnes deconvolution, reciprocal-sinc poles, and coefficient-threshold asymptotics.
Confluent digital extrapolation report	Reciprocal-moment Appell systems, Strang–Fix/cardinal reproduction, q -binomial scale closure, confluent Richardson filters, and Lambert-phase tomography.
Gamma-duality/total-positivity report	Laguerre–Pólya and PF_∞ structure, generalized gamma-convolution duality, Thorin measure, theta completion, mod-Gamma endpoint tilting, and large-order transfer.

Already developed in the corpus	Excluded from novelty claims here
Fourier-decay campaign	Exact zero orders, Thue–Morse lobe signs, sharp global envelopes, log-periodic fluctuations, and phase-sensitive real-axis decay.

Targeted searches were then made for the exact structures used below: Faber–Schauder expansions of F or G , Haar expansions from dyadic Fabius masses, Walsh–Hadamard expansions of p , finite-product Fourier transforms of Walsh modes, Bernstein beta-mixtures, and the sinc^4 Haar-energy expansion. No corresponding package was located in the audited TeX corpus. General Haar, Walsh, Schauder, and Bernstein theory is classical; the frontier content is the exact specialization and synthesis for the Fabius–Rvachev system.

1.3 Status ledger

ID	Statement	Status
A	Projective rational dyadic mass array and fixed-cell uniform-mixture representations, including locally uniform entire transform limits.	proved here
B	Rational unnormalized Haar and Faber–Schauder representations, with an orthonormal Haar/Fourier product expansion.	proved here
C	Walsh–Hadamard representation, reflection selection rule, and a series for $\widehat{\text{up}}$ whose terms are finite trigonometric products.	proved here
D	Exact Hamming-weight cancellation order, first nonzero Walsh moment, and $q = 1/2$ Gaussian-binomial aggregation.	proved here
E	Bell–Bernoulli formula for every higher Walsh moment.	proved here
F	Positive beta-density mixture and hypergeometric transform limits.	proved here
G	Equal-weight inverse-quantile chord mixtures and their finite sinc-product transform formula.	proved here
H	All-orders Haar-energy expansion generated by sinc^4 , with a superalgebraically small alias remainder.	proved here
I	Lambert-phase asymptotics for the boundary Schauder coefficients of $G = F^{-1}$ and boundary domination of the global interpolation error.	conjectural
J	A log-periodic transseries for the beyond-all-orders Haar alias remainder.	conjectural

2 Normalization and the function triad

2.1 Bounded Fabius function, density, and Rvachev function

Throughout, $F : \mathbb{R} \rightarrow [0, 1]$ is the bounded Fabius distribution function,

$$F(x) = 0 \quad (x \leq 0), \quad F(x) = 1 \quad (x \geq 1),$$

and up is Rvachev’s even C^∞ function supported on $[-1, 1]$. On their common fundamental intervals,

$$F(x) = \text{up}(x - 1), \quad 0 \leq x \leq 1, \quad (1)$$

$$F'(x) = 2 \text{up}(2x - 1) =: p(x), \quad 0 < x < 1, \quad (2)$$

$$F(x) + F(1 - x) = 1, \quad 0 \leq x \leq 1, \quad (3)$$

$$\text{up}(x) = 1 - F(x) = F(1 - x), \quad 0 \leq x \leq 1. \quad (4)$$

The density p , extended by zero outside $[0, 1]$, is a compactly supported C^∞ function satisfying

$$p(x) = p(1 - x), \quad \int_0^1 p(x) dx = 1. \quad (5)$$

The ordinary inverse of $F|_{[0,1]}$ is denoted

$$G = F^{-1} : [0, 1] \longrightarrow [0, 1], \quad G(1 - u) = 1 - G(u). \quad (6)$$

The signed global extension $\mathcal{F}$, important in the repository's repeated-integration theory, is not needed for the multiresolution proofs; it appears only in the baseline atlas.

2.2 Fourier convention

For an integrable function f we use angular frequency,

$$\widehat{f}(t) = \int_{\mathbb{R}} f(x) e^{-itx} dx, \quad f(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{f}(t) e^{itx} dt. \quad (7)$$

Set

$$\Phi(t) = \widehat{\text{up}}(t), \quad M(z) = \int_{-1}^1 \text{up}(x) e^{zx} dx = \Phi(iz). \quad (8)$$

The compact-interval Fourier image of the bounded Fabius function is

$$\mathcal{F}_F(t) = \int_0^1 F(x) e^{-itx} dx. \quad (9)$$

This notation avoids the meaningless full-line Fourier transform of a function that equals one on $[1, \infty)$.

2.3 Probabilistic model

Let $U_1, U_2, \dots$ be independent uniform random variables on $[-1, 1]$ and let $V_j = (U_j + 1)/2$, uniform on $[0, 1]$. Then

$$Y = \sum_{j \geq 1} 2^{-j} U_j, \quad X = \frac{Y + 1}{2} = \sum_{j \geq 1} 2^{-j} V_j. \quad (10)$$

The law of Y has density up , and F is the distribution function of X . Consequently

$$X \stackrel{d}{=} G(U), \quad Y \stackrel{d}{=} 2G(U) - 1, \quad U \sim \text{Unif}[0, 1]. \quad (11)$$

3 A representation atlas from the existing theory

This section collects a broad common baseline. Most formulas are already present, sometimes in stronger form, in the audited repository. They are included so that the genuinely new representations later can be compared against one normalization without forcing the reader to move among several manuscripts.

3.1 Infinite convolution, cube integrals, and finite heads

Let ν_a denote the uniform probability density on $[-a, a]$. Independence in (10) gives the infinite convolution

$$\text{up} = \nu_{1/2} * \nu_{1/4} * \nu_{1/8} * \dots \quad (12)$$

In distributional notation this is the infinite-cube integral

$$\text{up}(x) = \int_{[-1,1]^{\mathbb{N}}} \delta \left(x - \sum_{j \geq 1} 2^{-j} u_j \right) \prod_{j \geq 1} \frac{du_j}{2}, \quad (13)$$

and

$$F(x) = \int_{[0,1]^{\mathbb{N}}} \mathbf{1}_{\{\sum_{j \geq 1} 2^{-j} v_j \leq x\}} \prod_{j \geq 1} dv_j. \quad (14)$$

For the finite head

$$Y_N = \sum_{j=1}^N 2^{-j} U_j, \quad A_N = \sum_{j=1}^N 2^{-j} = 1 - 2^{-N},$$

write B_N for its density. Inclusion–exclusion for a box spline gives

$$B_N(x) = \frac{2^{N(N-1)/2}}{(N-1)!} \sum_{\varepsilon \in \{0,1\}^N} (-1)^{|\varepsilon|} \left(x + A_N - 2 \sum_{j=1}^N \varepsilon_j 2^{-j} \right)_+^{N-1}. \quad (15)$$

The signs $(-1)^{|\varepsilon|}$ are the finite Thue–Morse signs. Likewise, for $X_N = \sum_{j=1}^N 2^{-j} V_j$,

$$\mathbb{P}(X_N \leq x) = \frac{2^{N(N+1)/2}}{N!} \sum_{\varepsilon \in \{0,1\}^N} (-1)^{|\varepsilon|} \left(x - \sum_{j=1}^N \varepsilon_j 2^{-j} \right)_+^N. \quad (16)$$

Taking $N \rightarrow \infty$ gives $B_N \rightarrow \text{up}$ and the finite CDFs tend to F .

The tail is an independent scaled copy:

$$X \stackrel{d}{=} X_N + 2^{-N} X', \quad Y \stackrel{d}{=} Y_N + 2^{-N} Y'. \quad (17)$$

Hence the exact finite-scale identities

$$F(x) = \mathbb{E} F(2^N(x - X_N)), \quad (18)$$

$$\text{up}(x) = 2^N \int_{\mathbb{R}} B_N(y) \text{up}(2^N(x - y)) dy. \quad (19)$$

These formulas combine a finite Thue–Morse spline with a shrunken copy of the same limiting function.

3.2 Sinc, hyperbolic-sinc, and renormalization products

The Fourier and moment-generating products are

$$\Phi(t) = \prod_{j=1}^{\infty} \text{sinc} \left(\frac{t}{2^j} \right), \quad \text{sinc } z = \frac{\sin z}{z}, \quad (20)$$

$$M(z) = \prod_{j=1}^{\infty} \frac{\sinh(z/2^j)}{z/2^j}. \quad (21)$$

They converge locally uniformly on $\mathbb{C}$, and

$$\Phi(2t) = \text{sinc}(t) \Phi(t), \quad M(2z) = \frac{\sinh z}{z} M(z). \quad (22)$$

Iteration at any finite depth gives

$$\Phi(t) = \Phi(t/2^N) \prod_{j=1}^N \operatorname{sinc}(t/2^j), \quad (23)$$

which is the transform counterpart of (19).

Euler’s product for sine yields two canonical products:

$$\Phi(t) = \prod_{m=1}^{\infty} \left(\frac{t^2}{4\pi^2 m^2}; \frac{1}{4} \right)_{\infty}, \quad (24)$$

$$= \prod_{n=1}^{\infty} \left(1 - \frac{t^2}{4\pi^2 n^2} \right)^{1+v_2(n)}. \quad (25)$$

Thus $2\pi n$ is a zero of multiplicity $1 + v_2(n)$. For $|t| < 2\pi$,

$$\log \Phi(t) = - \sum_{r=1}^{\infty} \frac{\zeta(2r)}{r\pi^{2r}(4^r - 1)} t^{2r}. \quad (26)$$

The Wick-rotated determinant

$$\mathcal{A}(s) = \Phi(i\sqrt{s}) = \prod_{n=1}^{\infty} \left(1 + \frac{s}{4\pi^2 n^2} \right)^{1+v_2(n)} \quad (27)$$

underlies the spectral-zeta, total-positivity, and generalized-gamma-convolution results already developed in the corpus.

3.3 Moments, cumulants, Bell polynomials, and Bernoulli numbers

Write

$$\mu_n = \int_{-1}^1 x^n \operatorname{up}(x) \, dx = \mathbb{E}Y^n. \quad (28)$$

Odd moments vanish. The even cumulants are

$$\kappa_{2r}(Y) = \frac{2^{2r} B_{2r}}{2r(2^{2r} - 1)}, \quad \kappa_{2r+1}(Y) = 0. \quad (29)$$

Therefore

$$\mu_n = Y_n(0, \kappa_2, 0, \kappa_4, \dots, \kappa_n), \quad (30)$$

where Y_n is the complete exponential Bell polynomial, and

$$\Phi(t) = \sum_{r=0}^{\infty} \frac{(-1)^r \mu_{2r}}{(2r)!} t^{2r}. \quad (31)$$

For $m_n = \mathbb{E}X^n$ the smoothing equation $X \stackrel{d}{=} (V + X')/2$ gives

$$(2^n - 1)m_n = \sum_{k=0}^{n-1} \binom{n}{k} \frac{m_k}{n - k + 1}, \quad m_0 = 1. \quad (32)$$

All m_n are rational, and

$$\mu_{2r} = \frac{2m_{2r+1}}{2r + 1}. \quad (33)$$

3.4 Fourier inversion, twisted Poisson summation, and cosine series

Since Φ is rapidly decreasing on the real axis,

$$\text{up}(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \Phi(t) e^{itx} dt = \frac{1}{\pi} \int_0^\infty \Phi(t) \cos(tx) dt. \quad (34)$$

The distribution function of the centered law gives, for $0 \leq x \leq 1$,

$$F(x) = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \Phi(t) \frac{\sin((2x-1)t)}{t} dt. \quad (35)$$

Combining this with $F(x) = \text{up}(x-1)$ produces the integral identity

$$\int_0^\infty \Phi(t) \cos((1-x)t) dt = \frac{\pi}{2} + \int_0^\infty \Phi(t) \frac{\sin((2x-1)t)}{t} dt. \quad (36)$$

The partition of unity $\sum_{k \in \mathbb{Z}} \text{up}(x-k) = 1$ kills all nonzero integer Fourier modes. Twisting by $(-1)^k$ leaves the half-integer modes:

$$\sum_{k \in \mathbb{Z}} (-1)^k \text{up}(x-k) = \sum_{m \in \mathbb{Z}} \Phi((2m+1)\pi) e^{i(2m+1)\pi x}. \quad (37)$$

For $x \in [0, 1]$ the left side is $1 - 2F(x)$. With

$$A_m = \Phi((2m+1)\pi), \quad (38)$$

we obtain the rapidly convergent series

$$\text{up}(x) = \frac{1}{2} + \sum_{m=0}^\infty A_m \cos((2m+1)\pi x), \quad |x| \leq 1, \quad (39)$$

$$F(x) = \frac{1}{2} - \sum_{m=0}^\infty A_m \cos((2m+1)\pi x), \quad 0 \leq x \leq 1, \quad (40)$$

$$p(x) = \sum_{m=0}^\infty A_m (2m+1)\pi \sin((2m+1)\pi x). \quad (41)$$

Every derivative may be taken termwise.

3.5 Fourier images of p , F , and $1 - F$

The affine relation between p and up gives

$$\widehat{p}(t) = e^{-it/2} \Phi(t/2). \quad (42)$$

Integration by parts then yields the exact compact-interval transforms

$$\mathcal{F}_F(t) = \frac{e^{-it/2} \Phi(t/2) - e^{-it}}{it}, \quad (43)$$

$$\int_0^1 (1 - F(x)) e^{-itx} dx = \frac{1 - e^{-it/2} \Phi(t/2)}{it}. \quad (44)$$

The singularities at $t = 0$ are removable. After Wick rotation,

$$\int_0^1 e^{zx} F(x) dx = \frac{e^z - e^{z/2} M(z/2)}{z}, \quad (45)$$

$$\int_0^1 e^{zx} (1 - F(x)) dx = \frac{e^{z/2} M(z/2) - 1}{z}. \quad (46)$$

3.6 Quantile, Stieltjes, and correlation representations

Quantile transport (11) gives

$$\Phi(t) = \int_0^1 \exp(-it(2G(u) - 1)) \, du, \quad (47)$$

$$M(z) = \int_0^1 \exp(z(2G(u) - 1)) \, du, \quad (48)$$

$$\mu_n = \int_0^1 (2G(u) - 1)^n \, du. \quad (49)$$

For $z \notin [-1, 1]$, the Cauchy transform is

$$\mathcal{S}(z) = \int_{-1}^1 \frac{\text{up}(x)}{z - x} \, dx = \int_0^1 \frac{du}{z - (2G(u) - 1)}. \quad (50)$$

If $\Im z > 0$,

$$\mathcal{S}(z) = -i \int_0^\infty e^{itz} \Phi(t) \, dt. \quad (51)$$

The autocorrelation and convolution powers are

$$\int_{\mathbb{R}} \text{up}(x) \text{up}(x + y) \, dx = \frac{1}{2\pi} \int_{\mathbb{R}} \Phi(t)^2 e^{ity} \, dt, \quad (52)$$

$$\text{up}^{*r}(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \Phi(t)^r e^{itx} \, dt, \quad r \in \mathbb{N}. \quad (53)$$

3.7 Legendre and Chebyshev coefficient integrals

The repository contains a Fourier–Legendre expansion. Fourier inversion gives a useful transform-side form of its coefficients. If

$$\text{up}(x) = \sum_{n \geq 0} \lambda_n P_n(x), \quad \lambda_n = \frac{2n+1}{2} \int_{-1}^1 \text{up}(x) P_n(x) \, dx, \quad (54)$$

then odd coefficients vanish and, with j_n the spherical Bessel function,

$$\lambda_{2r} = \frac{4r+1}{\pi} (-1)^r \int_0^\infty \Phi(t) j_{2r}(t) \, dt. \quad (55)$$

Indeed, $\int_{-1}^1 e^{itx} P_n(x) \, dx = 2i^n j_n(t)$.

A parallel Chebyshev representation, apparently not stated in the audited corpus, is

$$\text{up}(x) = \frac{a_0}{2} + \sum_{n \geq 1} a_n T_n(x), \quad a_n = \frac{2}{\pi} \int_0^\pi \text{up}(\cos \theta) \cos(n\theta) \, d\theta. \quad (56)$$

The odd coefficients vanish and

$$a_{2r} = \frac{2(-1)^r}{\pi} \int_0^\infty \Phi(t) J_{2r}(t) \, dt. \quad (57)$$

The function $\theta \mapsto \text{up}(\cos \theta)$ is smooth and periodic, so the Chebyshev coefficients decrease faster than every power.

4 The projective dyadic mass array

4.1 Definition and exact algebra

Put

$$n_N = 2^N, \quad h_N = 2^{-N}, \quad I_{N,k} = [kh_N, (k+1)h_N), \quad (58)$$

with the last interval closed at 1, and define

$$\Delta_{N,k} = F((k+1)h_N) - F(kh_N), \quad 0 \leq k < n_N. \quad (59)$$

These are the probabilities that X lies in the dyadic cells.

Theorem 4.1 (Projective rational mass array). *For every $N \geq 0$ the vector $\Delta_N = (\Delta_{N,0}, \dots, \Delta_{N,n_N-1})$ has the following properties:*

$$\Delta_{N,k} \in \mathbb{Q}_{>0}, \quad (60)$$

$$\sum_{k=0}^{n_N-1} \Delta_{N,k} = 1, \quad (61)$$

$$\Delta_{N,n_N-1-k} = \Delta_{N,k}, \quad (62)$$

$$\Delta_{N,k} = \Delta_{N+1,2k} + \Delta_{N+1,2k+1}. \quad (63)$$

Consequently the mass polynomial

$$D_N(z) = \sum_{k=0}^{n_N-1} \Delta_{N,k} z^k \quad (64)$$

has rational coefficients, satisfies $D_N(1) = 1$, and is palindromic:

$$D_N(z) = z^{n_N-1} D_N(z^{-1}). \quad (65)$$

Proof. Rationality is the exact dyadic-value theorem from the repository. Positivity comes from strict increase of F on $(0, 1)$. The sum telescopes. Reflection (62) follows from $F(1-x) = 1 - F(x)$, and (63) is additivity after bisecting a cell. The polynomial statements are immediate. $\square$

The array is therefore a rational projective system of probability vectors. It is useful to regard it as the common discrete skeleton behind all subsequent representations. No approximation has yet been made.

4.2 Fixed-cell uniform mixtures

Define the dyadic conditional-expectation density

$$p_N(x) = n_N \Delta_{N,k}, \quad x \in I_{N,k}. \quad (66)$$

Equivalently,

$$p_N(x) dx = \sum_{k=0}^{n_N-1} \Delta_{N,k} \text{Unif}(I_{N,k}). \quad (67)$$

This is a finite mixture of equal-width uniform laws with variable rational weights. On the centered up scale, put

$$u_{N,k} = \frac{2k+1}{n_N} - 1. \quad (68)$$

The corresponding approximation is

$$u_{PN}(u) = \frac{1}{2} p_N\left(\frac{u+1}{2}\right) = \frac{n_N}{2} \Delta_{N,k}, \quad |u - u_{N,k}| < \frac{1}{n_N}. \quad (69)$$

Theorem 4.2 (Martingale and uniform convergence). *Let $\mathcal{D}_N$ be the dyadic sigma-algebra on $[0, 1]$. Then*

$$p_N = \mathbb{E}[p \mid \mathcal{D}_N]. \quad (70)$$

Moreover,

$$\|p_N - p\|_\infty \longrightarrow 0, \quad \|p_N - p\|_{L^q(0,1)} \longrightarrow 0 \quad (0 < q < \infty), \quad (71)$$

and $\text{up}_N \rightarrow \text{up}$ in the corresponding centered norms. If $p \in C^1$, then

$$\|p_N - p\|_\infty \leq h_N \|p'\|_\infty. \quad (72)$$

Proof. The average of p on $I_{N,k}$ is $h_N^{-1} \int_{I_{N,k}} p = n_N \Delta_{N,k}$, proving (70). Uniform continuity gives uniform convergence of cell averages; the displayed bound follows from the mean-value theorem. Finite-measure L^q convergence follows from uniform convergence. $\square$

Remark 4.3. Although p_N is only piecewise constant, all its coefficients are exact. It is not the finite-head spline B_N of (15): B_N is obtained by truncating the random series, whereas p_N is obtained by conditioning the limiting density on a spatial dyadic partition. The two approximations therefore probe different structures.

4.3 Rational exponential-polynomial limits for the transforms

The transform of (67) can be written directly in terms of the mass polynomial.

Theorem 4.4 (Primal dyadic transform representations). *For every $z \in \mathbb{C}$,*

$$\widehat{p}_N(z) = \frac{n_N(1 - e^{-iz/n_N})}{iz} D_N(e^{-iz/n_N}). \quad (73)$$

Consequently

$$\Phi_N^{\text{cell}}(t) := e^{it} \widehat{p}_N(2t) \quad (74)$$

$$= \text{sinc}(t/n_N) \sum_{k=0}^{n_N-1} \Delta_{N,k} \exp \left[-it \left(\frac{2k+1}{n_N} - 1 \right) \right]. \quad (75)$$

The limits

$$\widehat{p}(z) = \lim_{N \rightarrow \infty} \frac{n_N(1 - e^{-iz/n_N})}{iz} D_N(e^{-iz/n_N}), \quad (76)$$

$$\Phi(t) = \lim_{N \rightarrow \infty} \text{sinc}(t/n_N) \sum_{k=0}^{n_N-1} \Delta_{N,k} e^{-itu_{N,k}} \quad (77)$$

are locally uniform on $\mathbb{C}$. Thus the infinite sinc product is a locally uniform limit of rational exponential polynomials times one elementary sinc window.

Proof. Integrating p_N cell by cell gives

$$\int_{kh_N}^{(k+1)h_N} n_N \Delta_{N,k} e^{-izx} dx = \Delta_{N,k} \frac{n_N(1 - e^{-iz/n_N})}{iz} e^{-izk/n_N},$$

which sums to (73). The affine change from p to up gives (75). If $K \subset \mathbb{C}$ is compact, then

$$\sup_{z \in K} |\widehat{p}_N(z) - \widehat{p}(z)| \leq e^{\sup_{z \in K} |\Im z|} \|p_N - p\|_1,$$

so convergence is locally uniform. The same argument applies after the affine change of variables. $\square$

Deleting the sinc window gives the midpoint-atomic representation

$$\Phi(t) = \lim_{N \rightarrow \infty} \sum_{k=0}^{n_N-1} \Delta_{N,k} e^{-itu_{N,k}}. \quad (78)$$

The cell version is usually more accurate because it integrates the test function across each cell rather than concentrating the mass at its midpoint.

After Wick rotation one obtains the moment-generating limit

$$M(z) = \lim_{N \rightarrow \infty} \frac{\sinh(z/n_N)}{z/n_N} \sum_{k=0}^{n_N-1} \Delta_{N,k} \exp \left[z \left(\frac{2k+1}{n_N} - 1 \right) \right]. \quad (79)$$

Equations (77) and (79) are the same finite mixture on the imaginary and real axes.

4.4 A second-order quadrature principle

The conditional-expectation property gives better weak accuracy than the pointwise bound (72) suggests.

Proposition 4.5 (Cell-average quadrature). *If $p, g \in C^1[0, 1]$, then*

$$\left| \int_0^1 g(x)(p(x) - p_N(x)) dx \right| \leq h_N^2 \|g'\|_\infty \|p'\|_\infty. \quad (80)$$

In particular, on each compact set $|z| \leq R$,

$$\widehat{p}_N(z) - \widehat{p}(z) = \mathcal{O}_R(4^{-N}), \quad \Phi_N^{\text{cell}}(z) - \Phi(z) = \mathcal{O}_R(4^{-N}). \quad (81)$$

Proof. On a cell I choose any $x_I \in I$. Since $\int_I (p - p_N) = 0$,

$$\int_I g(p - p_N) = \int_I (g - g(x_I))(p - p_N).$$

Now $\text{osc}_I g \leq h_N \|g'\|_\infty$ and $\int_I |p - p_N| \leq h_N \text{osc}_I p \leq h_N^2 \|p'\|_\infty$. Summing $n_N = h_N^{-1}$ cells proves (80). For exponential test functions, their derivatives are uniformly bounded on compact subsets of $\mathbb{C}$. $\square$

The 4^{-N} behavior is visible in the numerical experiment of [section 13](#). The atomic midpoint formula is also second order for a smooth density and smooth test function, but the sinc-window formula has a clearer probabilistic interpretation: it is exactly the transform of the conditional-expectation density.

4.5 Moments, L^q norms, and discrete entropies

For $q > 0$, uniform convergence gives

$$\int_0^1 p(x)^q dx = \lim_{N \rightarrow \infty} n_N^{q-1} \sum_{k=0}^{n_N-1} \Delta_{N,k}^q. \quad (82)$$

For $q = 2$ this is an exact rational approximation to the collision integral,

$$\int_{-1}^1 \text{up}(x)^2 dx = \frac{1}{2} \lim_{N \rightarrow \infty} n_N \sum_{k=0}^{n_N-1} \Delta_{N,k}^2. \quad (83)$$

Differentiating (82) at $q = 1$ gives the Shannon asymptotic

$$-\sum_{k=0}^{n_N-1} \Delta_{N,k} \log \Delta_{N,k} = \log n_N - \int_0^1 p(x) \log p(x) dx + o(1). \quad (84)$$

Here $p \log p$ extends continuously by zero at the endpoints. Thus the continuous entropy of the Fabius density is the constant term in the entropy of its exact rational dyadic mass vectors.

For ordinary moments, integrating a monomial over each uniform cell gives

$$m_r = \int_0^1 x^r p(x) dx = \lim_{N \rightarrow \infty} \frac{1}{(r+1)n_N^r} \sum_{k=0}^{n_N-1} \Delta_{N,k} ((k+1)^{r+1} - k^{r+1}). \quad (85)$$

Every finite approximant is rational.

5 Haar and Faber–Schauder representations

5.1 Rational unnormalized Haar increments

Let

$$\eta_{n,k} = \mathbf{1}_{I_{n+1,2k}} - \mathbf{1}_{I_{n+1,2k+1}}, \quad 0 \leq k < 2^n, \quad (86)$$

and let $h_{n,k} = 2^{n/2} \eta_{n,k}$ be the orthonormal Haar functions. Define

$$\gamma_{n,k} = 2^n (\Delta_{n+1,2k} - \Delta_{n+1,2k+1}) \in \mathbb{Q}, \quad (87)$$

$$c_{n,k} = \langle p, h_{n,k} \rangle = 2^{n/2} (\Delta_{n+1,2k} - \Delta_{n+1,2k+1}) = 2^{-n/2} \gamma_{n,k}. \quad (88)$$

Thus the coefficients of the unnormalized Haar representation are rational, while the orthonormal coefficients lie in $2^{n/2} \mathbb{Q}$.

Theorem 5.1 (Exact Haar expansion). *For every $N \geq 1$,*

$$p_N(x) = 1 + \sum_{n=0}^{N-1} \sum_{k=0}^{2^n-1} \gamma_{n,k} \eta_{n,k}(x) = 1 + \sum_{n=0}^{N-1} \sum_{k=0}^{2^n-1} c_{n,k} h_{n,k}(x). \quad (89)$$

The power-of-two partial sums converge uniformly and in every finite L^q :

$$p(x) = 1 + \lim_{N \rightarrow \infty} \sum_{n=0}^{N-1} \sum_{k=0}^{2^n-1} \gamma_{n,k} \eta_{n,k}(x). \quad (90)$$

Reflection gives

$$\gamma_{n,2^n-1-k} = -\gamma_{n,k}, \quad c_{n,2^n-1-k} = -c_{n,k}. \quad (91)$$

Proof. The difference $p_{n+1} - p_n$ is constant with opposite signs on the two children of each parent cell. On the left child it equals $2^n (\Delta_{n+1,2k} - \Delta_{n+1,2k+1}) = \gamma_{n,k}$, so $p_{n+1} - p_n = \sum_k \gamma_{n,k} \eta_{n,k}$. Telescoping from $p_0 \equiv 1$ proves (89). Convergence is [theorem 4.2](#); antisymmetry follows from (62). $\square$

The coefficients can also be read directly from three dyadic values:

$$c_{n,k} = 2^{n/2} \left[2F\left(\frac{2k+1}{2^{n+1}}\right) - F\left(\frac{k}{2^n}\right) - F\left(\frac{k+1}{2^n}\right) \right]. \quad (92)$$

This is a discrete second derivative of F at a dyadic midpoint.

5.2 Fourier image of the Haar expansion

A direct integration gives

$$\widehat{h}_{n,k}(z) = 2^{n/2} e^{-izk/2^n} \frac{(1 - e^{-iz/2^{n+1}})^2}{iz}. \quad (93)$$

Therefore:

Theorem 5.2 (Haar product-series transform). *Locally uniformly on $\mathbb{C}$,*

$$\widehat{p}(z) = \frac{1 - e^{-iz}}{iz} + \sum_{n=0}^{\infty} \sum_{k=0}^{2^n-1} c_{n,k} 2^{n/2} e^{-izk/2^n} \frac{(1 - e^{-iz/2^{n+1}})^2}{iz}. \quad (94)$$

Equivalently,

$$\Phi(t) = \text{sinc}(t) + e^{it} \sum_{n=0}^{\infty} \sum_{k=0}^{2^n-1} c_{n,k} 2^{n/2} e^{-2itk/2^n} \frac{(1 - e^{-2it/2^{n+1}})^2}{2it}. \quad (95)$$

Proof. Transform the finite identity (89) and let $N \rightarrow \infty$. Local uniform convergence follows from uniform L^1 convergence of p_N to p . Absolute convergence by scale also follows directly: smoothness gives $c_{n,k} = \mathcal{O}(2^{-3n/2})$ uniformly in k , while $\widehat{h}_{n,k}(z) = \mathcal{O}_K(2^{-3n/2})$ on a compact K ; summing 2^n locations leaves a convergent $\mathcal{O}_K(4^{-n})$ scale contribution. $\square$

The factor $(1 - e^{-iz/2^{n+1}})^2$ is the frequency signature of one vanishing moment. In section 6 it will be replaced by a product with as many sine factors as there are nonzero binary digits in the Walsh index.

5.3 Faber–Schauder series of the Fabius function

Let $\psi_{n,k}$ be the peak-one tent supported on $I_{n,k}$, with apex at

$$m_{n,k} = \frac{2k+1}{2^{n+1}}. \quad (96)$$

Define

$$d_{n,k} = F(m_{n,k}) - \frac{1}{2}F(k2^{-n}) - \frac{1}{2}F((k+1)2^{-n}) \quad (97)$$

$$= \frac{1}{2}(\Delta_{n+1,2k} - \Delta_{n+1,2k+1}) = 2^{-n/2-1} c_{n,k} \in \mathbb{Q}. \quad (98)$$

Theorem 5.3 (Rational Faber–Schauder representation). *The piecewise-linear interpolant $L_N F$ through the dyadic values $F(k2^{-N})$ is*

$$L_N F(x) = x + \sum_{n=0}^{N-1} \sum_{k=0}^{2^n-1} d_{n,k} \psi_{n,k}(x). \quad (99)$$

It converges uniformly to F , hence

$$F(x) = x + \sum_{n=0}^{\infty} \sum_{k=0}^{2^n-1} d_{n,k} \psi_{n,k}(x), \quad (100)$$

where the series is understood by complete dyadic levels. The coefficients are rational and obey

$$d_{n,2^n-1-k} = -d_{n,k}. \quad (101)$$

Moreover,

$$\|F - L_N F\|_{\infty} \leq \frac{\|F''\|_{\infty}}{8} 4^{-N}. \quad (102)$$

Proof. The classical Faber–Schauder construction says that the coefficient added at a new midpoint is its value minus the chord average, giving (97). Equation (98) follows by expanding the two child masses. Uniform convergence follows from continuity; the displayed error is the standard linear-interpolation bound on intervals of length 2^{-N} . Symmetry follows from $F(1-x) = 1 - F(x)$. $\square$

Because F is convex on $[0, 1/2]$ and concave on $[1/2, 1]$, one also has

$$d_{n,k} < 0 \quad (0 \leq k < 2^{n-1}), \quad d_{n,k} > 0 \quad (2^{n-1} \leq k < 2^n), \quad (103)$$

with the $n = 0$ coefficient zero by symmetry.

5.4 A sinc-squared series for the Fourier image of F

The transform of a peak-one tent of width 2^{-n} is

$$\int_0^1 \psi_{n,k}(x) e^{-izx} dx = \frac{2^{-n}}{2} e^{-izm_{n,k}} \operatorname{sinc}^2\left(\frac{z}{2^{n+2}}\right). \quad (104)$$

Since

$$\int_0^1 x e^{-izx} dx = \frac{e^{-iz}(1+iz) - 1}{z^2}, \quad (105)$$

we obtain a representation of the Fourier image of the Fabius function itself.

Corollary 5.4 (Schauder–sinc representation of $\mathcal{F}_F$). *Locally uniformly on $\mathbb{C}$,*

$$\mathcal{F}_F(z) = \frac{e^{-iz}(1+iz) - 1}{z^2} + \sum_{n=0}^{\infty} \sum_{k=0}^{2^n-1} d_{n,k} 2^{-n-1} e^{-izm_{n,k}} \operatorname{sinc}^2\left(\frac{z}{2^{n+2}}\right). \quad (106)$$

All coefficients $d_{n,k}$ are rational. Equation (106) is identical, after summation, to the compact product formula (43).

The identity equates a single infinite sinc product to a doubly indexed sum of translated sinc^2 atoms. It is a time–frequency multiresolution counterpart of the canonical zero product (25).

6 Walsh–Hadamard and Thue–Morse product representations

6.1 Paley Walsh functions and the finite Hadamard transform

Let

$$r_j(x) = (-1)^{\lfloor 2^{j+1}x \rfloor}, \quad j \geq 0, \quad (107)$$

away from dyadic endpoints. If

$$m = \sum_{j \geq 0} b_j 2^j, \quad b_j \in \{0, 1\}, \quad (108)$$

then the Paley Walsh function is

$$w_m(x) = \prod_{j \geq 0} r_j(x)^{b_j}. \quad (109)$$

Write

$$r(m) = \operatorname{wt}_2(m) = \sum_j b_j, \quad \sigma(m) = \sum_j j b_j. \quad (110)$$

The top mode w_{2^N-1} on a depth- N partition is the signed Thue–Morse step function.

Define the Walsh coefficient of the Fabius density by

$$a_m = \int_0^1 p(x) w_m(x) \, dx. \quad (111)$$

If $m < n_N$, then w_m is constant on every depth- N cell. With

$$\varepsilon_{m,N}(k) = w_m\left(\frac{k+1/2}{n_N}\right) \in \{\pm 1\}, \quad (112)$$

we have the exact finite formula

$$a_m = \sum_{k=0}^{n_N-1} \varepsilon_{m,N}(k) \Delta_{N,k}. \quad (113)$$

Thus every a_m is rational.

Let $H_N = (\varepsilon_{m,N}(k))_{0 \leq m, k < n_N}$. In vector notation,

$$a^{(N)} = H_N \Delta_N, \quad \Delta_N = n_N^{-1} H_N a^{(N)}, \quad H_N H_N^T = n_N I. \quad (114)$$

This is an exact rational Walsh–Hadamard transform of the dyadic mass vector.

Theorem 6.1 (Walsh expansion and finite Parseval). *For every N ,*

$$p_N(x) = \sum_{m=0}^{n_N-1} a_m w_m(x). \quad (115)$$

Consequently the power-of-two Walsh partial sums converge uniformly:

$$p(x) = \lim_{N \rightarrow \infty} \sum_{m=0}^{n_N-1} a_m w_m(x). \quad (116)$$

At each finite level,

$$\sum_{m=0}^{n_N-1} a_m^2 = n_N \sum_{k=0}^{n_N-1} \Delta_{N,k}^2 = \|p_N\|_2^2. \quad (117)$$

Hence

$$\|p\|_2^2 = \sum_{m=0}^{\infty} a_m^2, \quad \| \text{up} \|_2^2 = \frac{1}{2} \sum_{m=0}^{\infty} a_m^2. \quad (118)$$

Proof. On a depth- N cell, (115) is exactly the inverse Hadamard formula in (114). Uniform convergence is [theorem 4.2](#). Parseval follows from orthonormality of the Walsh functions or directly from $H_N H_N^T = n_N I$. $\square$

6.2 Reflection selection rules

Binary reflection complements every Rademacher sign, modulo a null set:

$$w_m(1-x) = (-1)^{r(m)} w_m(x). \quad (119)$$

Since $p(1-x) = p(x)$, this gives a sharp parity filter.

Proposition 6.2 (Hamming-parity selection). *The density coefficients satisfy*

$$a_m = 0 \quad \text{whenever } r(m) \text{ is odd.} \quad (120)$$

If

$$b_m = \int_0^1 \left(F(x) - \frac{1}{2} \right) w_m(x) \, dx, \quad (121)$$

then

$$b_m = 0 \quad \text{whenever } r(m) \text{ is even.} \quad (122)$$

Proof. Substitute $x \mapsto 1 - x$. For a_m the integral is multiplied by $(-1)^{r(m)}$; for b_m , the antisymmetry $F(1 - x) - 1/2 = -(F(x) - 1/2)$ contributes one additional minus sign. $\square$

Thus the Walsh support of the density occupies only even Hamming layers, while the centered distribution function occupies only odd layers. This is a digital analogue of the disappearance of odd ordinary Fourier and polynomial moments under reflection symmetry.

6.3 Finite-product Fourier transform of a Walsh mode

Set

$$W_m(z) = \int_0^1 w_m(x) e^{-izx} dx. \quad (123)$$

For $m > 0$, let

$$N(m) = 1 + \lfloor \log_2 m \rfloor. \quad (124)$$

Theorem 6.3 (Walsh finite-product transform). *For $m = 0$,*

$$W_0(z) = e^{-iz/2} \frac{2 \sin(z/2)}{z}. \quad (125)$$

For $m > 0$, with $N = N(m)$ and $r = r(m)$,

$$W_m(z) = \frac{1 - e^{-iz/2^N}}{iz} \prod_{j=0}^{N-1} \left(1 + (-1)^{b_j} e^{-iz/2^{j+1}} \right) \quad (126)$$

$$= e^{-iz/2} \frac{2^{N+1} i^r}{z} \sin\left(\frac{z}{2^{N+1}}\right) \prod_{b_j=0} \cos\left(\frac{z}{2^{j+2}}\right) \prod_{b_j=1} \sin\left(\frac{z}{2^{j+2}}\right). \quad (127)$$

Both right sides have removable singularities and define the same entire function.

Proof. At depth N , integrate the constant value of w_m on each cell. The common cell integral is $(1 - e^{-iz/2^N})/(iz)$, and the signed geometric sum factorizes according to the binary digits of the cell index:

$$\sum_{k=0}^{2^N-1} \varepsilon_{m,N}(k) q^k = \prod_{j=0}^{N-1} \left(1 + (-1)^{b_j} q^{2^{N-1-j}} \right), \quad q = e^{-iz/2^N}.$$

This proves (126). Pairing exponentials into sines and cosines gives (127). $\square$

6.4 A series of finite products for the infinite sinc product

Transforming the Walsh partial sums gives a representation of Φ that is qualitatively different from the canonical product: the coefficients encode Fabius arithmetic, while each basis transform is a finite product.

Theorem 6.4 (Walsh product-series for Φ). *Locally uniformly on $\mathbb{C}$,*

$$\Phi(t) = e^{it} \sum_{m=0}^{\infty} a_m W_m(2t), \quad (128)$$

where the sum is taken through complete blocks $m < 2^N$. More explicitly,

$$\begin{aligned} \Phi(t) = \operatorname{sinc}(t) + \sum_{\substack{m \geq 1 \\ r(m) \text{ even}}} a_m \frac{2^{N(m)} (-1)^{r(m)/2}}{t} \sin\left(\frac{t}{2^{N(m)}}\right) \\ \times \prod_{b_j=0} \cos\left(\frac{t}{2^{j+1}}\right) \prod_{b_j=1} \sin\left(\frac{t}{2^{j+1}}\right). \end{aligned} \quad (129)$$

Each coefficient a_m is rational and is given exactly by (113).

Proof. Since $p_N \rightarrow p$ in L^1 and all functions are supported in $[0, 1]$, their Fourier transforms converge locally uniformly. Substitute the finite Walsh expansion (115) and use $\Phi(t) = e^{it}\widehat{p}(2t)$. Equation (129) follows from theorem 6.3 and the parity filter (120). $\square$

The corresponding hyperbolic product-series is obtained by $t \mapsto iz$:

$$M(z) = e^{-z} \sum_{m=0}^{\infty} a_m W_m(2iz), \quad (130)$$

so each term becomes a finite product of sinh and cosh factors. Equations (20) and (129) therefore give a product-versus-product-series duality for the same entire function.

6.5 Exact Hamming-weight cancellation

The number of sine factors in (127) is not cosmetic: it is the exact number of vanishing algebraic moments.

Theorem 6.5 (Walsh cancellation order and first moment). *For $m > 0$, put $r = r(m)$ and $\sigma = \sigma(m)$. Then*

$$\int_0^1 x^q w_m(x) dx = 0, \quad 0 \leq q < r, \quad (131)$$

and

$$\int_0^1 x^r w_m(x) dx = (-1)^r r! 2^{-2r-\sigma}. \quad (132)$$

Thus W_m has a zero of exact order $r(m)$ at the origin.

Proof. Extract the small factors from (127). One obtains

$$W_m(z) = i^r 2^{-2r-\sigma} z^r R_m(z), \quad R_m(0) = 1, \quad (133)$$

where R_m is entire. On the other hand,

$$W_m(z) = \sum_{q=0}^{\infty} \frac{(-iz)^q}{q!} \int_0^1 x^q w_m(x) dx. \quad (134)$$

Comparing the first nonzero Taylor coefficient proves both statements. $\square$

For the top Thue–Morse mode, $m = 2^N - 1$, one has $r = N$ and $\sigma = N(N-1)/2$. Hence

$$\int_0^1 x^q w_{2^N-1}(x) dx = 0 \quad (q < N), \quad \int_0^1 x^N w_{2^N-1}(x) dx = (-1)^N N! 2^{-N(N+3)/2}. \quad (135)$$

This is the continuous, normalized Prouhet cancellation attached to the Thue–Morse sign pattern.

6.6 The Gaussian-binomial cancellation spectrum

The first nonzero moments aggregate exactly by Hamming weight.

Theorem 6.6 ($q = 1/2$ Gaussian-binomial aggregation). *For $0 \leq r \leq N$,*

$$\sum_{\substack{0 \leq m < 2^N \\ r(m)=r}} 2^{-\sigma(m)} = 2^{-r(r-1)/2} \begin{bmatrix} N \\ r \end{bmatrix}_{1/2}. \quad (136)$$

Consequently

$$\sum_{\substack{0 \leq m < 2^N \\ r(m)=r}} \int_0^1 x^r w_m(x) dx = (-1)^r r! 2^{-2r-r(r-1)/2} \begin{bmatrix} N \\ r \end{bmatrix}_{1/2}. \quad (137)$$

Proof. The finite q -binomial theorem gives

$$\prod_{j=0}^{N-1} (1 + y2^{-j}) = \sum_{r=0}^N 2^{-r(r-1)/2} \begin{bmatrix} N \\ r \end{bmatrix}_{1/2} y^r.$$

The coefficient of y^r on the left is the sum of $2^{-\sigma(m)}$ over all subsets of r bit positions. Combine with (132). $\square$

The appearance of $q = 1/2$, rather than the $q = 1/4$ common in the even-moment and scale-energy theory, has a simple origin: it records bit *positions*, not squared dyadic lengths. Squaring amplitudes or passing to even cumulants naturally changes $1/2$ to $1/4$.

6.7 All higher Walsh moments: a Bell–Bernoulli formula

The product formula encodes every moment, not only the first nonzero one. Define the even entire factor

$$E_m(z) = \operatorname{sinc}\left(\frac{z}{2^{N+1}}\right) \prod_{b_j=0} \cos\left(\frac{z}{2^{j+2}}\right) \prod_{b_j=1} \operatorname{sinc}\left(\frac{z}{2^{j+2}}\right), \quad (138)$$

where $N = N(m)$. Then

$$R_m(z) = e^{-iz/2} E_m(z) \quad (139)$$

in (133). Write

$$E_m(z) = \sum_{j=0}^{\infty} e_{m,j} z^{2j}. \quad (140)$$

For $q \geq 1$, set

$$\lambda_{2q}(m) = -\frac{2^{2q-1}|B_{2q}|}{q} \left[2^{-2q(N+1)} + \sum_{b_j=1} 2^{-2q(j+2)} + (2^{2q} - 1) \sum_{b_j=0} 2^{-2q(j+2)} \right]. \quad (141)$$

The classical expansions of $\log \operatorname{sinc}$ and $\log \cos$ show that

$$\log E_m(z) = \sum_{q=1}^{\infty} \lambda_{2q}(m) \frac{z^{2q}}{(2q)!}. \quad (142)$$

Therefore

$$e_{m,j} = \frac{1}{(2j)!} Y_{2j}(0, \lambda_2(m), 0, \lambda_4(m), \dots, \lambda_{2j}(m)). \quad (143)$$

Theorem 6.7 (Bell–Bernoulli formula for Walsh moments). *Let $m > 0$, $r = r(m)$, and $\ell \geq 0$. Then*

$$\int_0^1 x^{r+\ell} w_m(x) dx = (-1)^r \frac{(r+\ell)!}{2^{2r+\sigma(m)}} \sum_{j=0}^{\lfloor \ell/2 \rfloor} \frac{(-1)^j e_{m,j}}{2^{\ell-2j}(\ell-2j)!}, \quad (144)$$

with $e_{m,j}$ given explicitly by (143) and (141).

Proof. Use $W_m(z) = i^r 2^{-2r-\sigma} z^r e^{-iz/2} E_m(z)$, expand the exponential and the even series (140), and compare with (134). Formula (143) is the standard exponential Bell-polynomial coefficient formula applied to (142). $\square$

This theorem places the Bernoulli numbers in the *basis moments*, while the rational coefficients a_m carry the Fabius-specific arithmetic. The ordinary Fabius moments can therefore be reconstructed as

$$m_q = \lim_{N \rightarrow \infty} \sum_{m=0}^{2^N-1} a_m \int_0^1 x^q w_m(x) dx. \quad (145)$$

Only modes with even $r(m) \leq q$ contribute. For example, since $m_2 = 5/18$,

$$\sum_{\substack{m \geq 1 \\ r(m)=2}} 2^{-\sigma(m)} a_m = -\frac{4}{9}. \quad (146)$$

Indeed, the constant Walsh mode contributes $\int_0^1 x^2 dx = 1/3$, and (132) contributes $2^{-3-\sigma(m)} a_m$ from every weight-two mode.

Remark 6.8. Equation (145) is naturally ordered by dyadic blocks, not by arbitrary rearrangement. Blockwise convergence follows immediately from uniform convergence of the Walsh conditional expectations. Establishing absolute convergence after regrouping by Hamming weight is a separate and interesting coefficient-decay problem.

7 Bernstein polynomials and positive beta-mixture limits

7.1 Rational Bernstein approximants

For $n = n_N = 2^N$, define the Bernstein polynomial

$$\mathcal{B}_N F(x) = \sum_{k=0}^n F(k/n) \binom{n}{k} x^k (1-x)^{n-k}. \quad (147)$$

Every coefficient in the Bernstein basis is a rational dyadic Fabius value. The power-basis form is also rational:

$$\mathcal{B}_N F(x) = \sum_{j=0}^n \binom{n}{j} \left(\sum_{k=0}^j (-1)^{j-k} \binom{j}{k} F(k/n) \right) x^j. \quad (148)$$

Thus

$$F(x) = \lim_{N \rightarrow \infty} \mathcal{B}_N F(x) \quad (149)$$

uniformly, as a limit of explicit rational polynomials.

The dyadic Faber interpolants of [section 5](#) converge at order 4^{-N} because they exploit the C^2 smoothness on a mesh of width 2^{-N} . The Bernstein approximants use a binomial cloud of width $n^{-1/2}$ and have the classical expansion

$$\mathcal{B}_N F(x) - F(x) = \frac{x(1-x)}{2n} F''(x) + \mathcal{O}(n^{-2}) \quad (150)$$

under the available smoothness. Their advantage is positivity and the probabilistic mixture structure of the derivative.

7.2 Derivative as a beta-density mixture

Differentiating (147) gives

$$p_N^B(x) := (\mathcal{B}_N F)'(x) \quad (151)$$

$$= n \sum_{k=0}^{n-1} \Delta_{N,k} \binom{n-1}{k} x^k (1-x)^{n-1-k} \quad (152)$$

$$= \sum_{k=0}^{n-1} \Delta_{N,k} \frac{x^k (1-x)^{n-1-k}}{B(k+1, n-k)}. \quad (153)$$

Theorem 7.1 (Positive rational beta-mixture representation). *The functions p_N^B are probability densities and are finite mixtures of $\text{Beta}(k+1, n-k)$ densities with the exact rational weights $\Delta_{N,k}$. Moreover,*

$$p_N^B \longrightarrow p \quad \text{uniformly on } [0, 1]. \quad (154)$$

Consequently p is a uniform limit of positive polynomial densities whose mixture weights are exactly the dyadic Fabius masses.

Proof. Formula (153) proves positivity and unit mass. The classical derivative theorem for Bernstein polynomials states that $(\mathcal{B}_n f)' \rightarrow f'$ uniformly when $f \in C^1[0, 1]$. $\square$

This representation is structurally different from the uniform-cell mixture:

model	component family	weights
p_N	disjoint equal-width uniforms	$\Delta_{N,k}$
p_N^B	overlapping beta densities	$\Delta_{N,k}$

The same rational vector is therefore a mixing law for both a local and a global positive basis.

7.3 Hypergeometric transform representations

The moment-generating function of a $\text{Beta}(a, b)$ variable is ${}_1F_1(a; a+b; z)$. Hence:

Theorem 7.2 (Confluent-hypergeometric mixture limits). *Locally uniformly on $\mathbb{C}$,*

$$\int_0^1 p(x) e^{zx} dx = \lim_{N \rightarrow \infty} \sum_{k=0}^{n_N-1} \Delta_{N,k} {}_1F_1(k+1; n_N+1; z), \quad (155)$$

$$\Phi(t) = e^{it} \lim_{N \rightarrow \infty} \sum_{k=0}^{n_N-1} \Delta_{N,k} {}_1F_1(k+1; n_N+1; -2it), \quad (156)$$

$$M(z) = e^{-z} \lim_{N \rightarrow \infty} \sum_{k=0}^{n_N-1} \Delta_{N,k} {}_1F_1(k+1; n_N+1; 2z). \quad (157)$$

Thus the infinite sinc and hyperbolic-sinc products are limits of positive rational mixtures of confluent hypergeometric functions.

Proof. Integrate (153) against e^{zx} and use uniform convergence. The affine transform relations $\Phi(t) = e^{it} \hat{p}(2t)$ and $M(z) = e^{-z} \int_0^1 p(x) e^{2zx} dx$ give the remaining formulas. $\square$

The beta moments give another rational moment limit:

$$m_r = \lim_{N \rightarrow \infty} \sum_{k=0}^{n_N-1} \Delta_{N,k} \frac{(k+1)_r}{(n_N+1)_r}, \quad r \in \mathbb{N}, \quad (158)$$

where $(a)_r$ is the rising factorial. Comparison of (158) with the cell formula (85) gives two positive rational quadrature schemes for the same exact Fabius moments.

8 Dual quantile cells and equal-weight sinc mixtures

8.1 The inverse dyadic width array

The primal mass array fixes spatial widths and varies probabilities. The inverse construction fixes probabilities and varies spatial widths. Put

$$g_{N,k} = G(k/n_N), \quad q_{N,k} = g_{N,k+1} - g_{N,k} > 0, \quad 0 \leq k < n_N. \quad (159)$$

Then

$$\sum_{k=0}^{n_N-1} q_{N,k} = 1, \quad (160)$$

$$q_{N,n_N-1-k} = q_{N,k}, \quad (161)$$

$$q_{N,k} = q_{N+1,2k} + q_{N+1,2k+1}. \quad (162)$$

Unlike $\Delta_{N,k}$, these widths are not expected to be rational in general. They are exact inverse-Fabius data.

Let $G_N = L_N G$ be the piecewise-linear interpolant through the points $(k/n_N, g_{N,k})$. Pushing uniform measure forward by G_N gives the density

$$p_N^Q(x) = \frac{1}{n_N q_{N,k}}, \quad g_{N,k} < x < g_{N,k+1}. \quad (163)$$

Each interval has probability $1/n_N$. Thus p_N^Q is a finite mixture of variable-width uniforms with *equal* weights.

Primal–dual cell dictionary

	spatial interval	probability weight
primal Fabius cells	fixed width $1/n_N$	variable $\Delta_{N,k}$
dual quantile cells	variable width $q_{N,k}$	fixed $1/n_N$

The two projective arrays are interchanged by F and G : the first samples increments of F on a uniform spatial grid, the second samples increments of G on a uniform probability grid.

8.2 Exact chord integration and the Fourier image

On the k th probability cell, G_N moves linearly from $g_{N,k}$ to $g_{N,k+1}$. Integrating an exponential along that chord gives a sinc factor.

Theorem 8.1 (Inverse-quantile chord representation). *Locally uniformly on $\mathbb{C}$,*

$$\Phi(t) = \lim_{N \rightarrow \infty} \frac{1}{n_N} \sum_{k=0}^{n_N-1} e^{-it(g_{N,k} + g_{N,k+1} - 1)} \operatorname{sinc}(t q_{N,k}), \quad (164)$$

$$M(z) = \lim_{N \rightarrow \infty} \frac{1}{n_N} \sum_{k=0}^{n_N-1} e^{z(g_{N,k} + g_{N,k+1} - 1)} \frac{\sinh(z q_{N,k})}{z q_{N,k}}. \quad (165)$$

Equivalently, Φ is the locally uniform limit of the Fourier transforms of finite equal-weight uniform mixtures on the centered intervals

$$[2g_{N,k} - 1, 2g_{N,k+1} - 1]. \quad (166)$$

Proof. Uniform continuity of G gives $G_N \rightarrow G$ uniformly. Therefore

$$\int_0^1 e^{-it(2G_N(u)-1)} du \longrightarrow \int_0^1 e^{-it(2G(u)-1)} du = \Phi(t)$$

locally uniformly in t . On a probability cell of length $1/n_N$, direct integration of the affine phase gives

$$\frac{1}{n_N} e^{-it(g_{N,k}+g_{N,k+1}-1)} \operatorname{sinc}(tq_{N,k}).$$

Summing proves (164); Wick rotation gives (165). $\square$

Removing the chord average gives the simpler midpoint Riemann representation

$$\Phi(t) = \lim_{N \rightarrow \infty} \frac{1}{n_N} \sum_{k=0}^{n_N-1} \exp \left[-it \left(2G \left(\frac{k+1/2}{n_N} \right) - 1 \right) \right]. \quad (167)$$

The chord formula is preferable when inverse values at endpoints are already available, because it integrates the linear interpolant exactly.

8.3 Moment and Laplace limits from inverse dyadic values

For the uncentered law, exact integration of powers along each chord yields:

Corollary 8.2 (Inverse-dyadic moment polygons). *For every integer $r \geq 0$,*

$$m_r = \lim_{N \rightarrow \infty} \frac{1}{n_N(r+1)} \sum_{k=0}^{n_N-1} \frac{g_{N,k+1}^{r+1} - g_{N,k}^{r+1}}{g_{N,k+1} - g_{N,k}} \quad (168)$$

$$= \lim_{N \rightarrow \infty} \frac{1}{n_N(r+1)} \sum_{k=0}^{n_N-1} \sum_{j=0}^r g_{N,k}^j g_{N,k+1}^{r-j}. \quad (169)$$

For arbitrary $z \in \mathbb{C}$,

$$\int_0^1 p(x) e^{zx} dx = \lim_{N \rightarrow \infty} \frac{1}{n_N} \sum_{k=0}^{n_N-1} e^{z(g_{N,k}+g_{N,k+1})/2} \operatorname{sinhc} \left(\frac{zq_{N,k}}{2} \right), \quad (170)$$

where $\operatorname{sinhc} w = \sinh(w)/w$.

The first line is a limit of complete homogeneous symmetric polynomials in adjacent inverse-dyadic values. It complements the repository's local inverse series and exact inverse values at special dyadic levels: local inverse arithmetic enters a global moment formula through a simple polygonal limit.

8.4 Dual L^q spectra and entropy constants

Because $q_{N,k} = \int_{k/n_N}^{(k+1)/n_N} G'(u) du$, the slope $n_N q_{N,k}$ is the dyadic conditional expectation of G' . This gives a discrete counterpart of the inverse-derivative norm threshold.

Theorem 8.3 (Primal–dual Rényi spectra). *For $s > 0$,*

$$n_N^{s-1} \sum_k \Delta_{N,k}^s \longrightarrow \int_0^1 p(x)^s dx. \quad (171)$$

For $0 < s < 1$,

$$n_N^{s-1} \sum_k q_{N,k}^s \longrightarrow \int_0^1 G'(u)^s du = \int_0^1 p(x)^{1-s} dx. \quad (172)$$

At $s = 1$ both normalized sums equal one. For $s > 1$, the dual normalized sum tends to $+\infty$, consistently with $G' \notin L^s(0, 1)$.

Proof. The primal statement is (82). On each probability cell, $n_N q_{N,k} = \mathbb{E}[G' \mid \mathcal{D}_N]$. The dyadic martingale converges to G' in L^1 . For $0 < s < 1$, the inequality $|a^s - b^s| \leq |a - b|^s$ and Hölder's inequality give convergence of the s th powers in L^1 . The change of variables $u = F(x)$ gives

$$\int_0^1 G'(u)^s du = \int_0^1 p(x)^{1-s} dx.$$

For $s > 1$, conditional-expectation L^s norms increase to the infinite L^s norm of G' . $\square$

The one-sided derivatives at $s = 1$ give entropy constants. In addition to (84), the endpoint asymptotics imply $\log p \in L^1(0, 1)$, and

$$-\sum_{k=0}^{n_N-1} q_{N,k} \log q_{N,k} = \log n_N + \int_0^1 \log p(x) dx + o(1). \quad (173)$$

Thus the two exact partitions retain different continuous entropies:

$$H(\Delta_N) - \log n_N \longrightarrow -\int p \log p, \quad (174)$$

$$H(q_N) - \log n_N \longrightarrow \int \log p. \quad (175)$$

The second constant is sensitive to endpoint flatness, whereas the first is weighted by the probability density itself.

9 Haar scale energies and the sinc⁴ continuum expansion

9.1 Three exact forms of the energy

Define the detail energy

$$E_n = \|p_{n+1} - p_n\|_2^2 = \sum_{k=0}^{2^n-1} c_{n,k}^2. \quad (176)$$

Using (88),

$$E_n = 2^n \sum_{k=0}^{2^n-1} (\Delta_{n+1,2k} - \Delta_{n+1,2k+1})^2 \in \mathbb{Q}. \quad (177)$$

Thus a sum of squares of generally quadratic Haar coefficients is rational. Orthogonality gives

$$\|p_N\|_2^2 = 1 + \sum_{n=0}^{N-1} E_n, \quad \|p\|_2^2 = 1 + \sum_{n=0}^{\infty} E_n. \quad (178)$$

The Haar and Walsh filtrations are the same nested dyadic subspaces. Hence:

Proposition 9.1 (Haar–Walsh shell identity). *For every $n \geq 0$,*

$$E_n = \sum_{m=2^n}^{2^{n+1}-1} a_m^2. \quad (179)$$

Only even-Hamming-weight indices contribute to the right side.

Proof. Both sides are the squared norm of the orthogonal projection of p onto $V_{n+1} \ominus V_n$, where V_n is the space of functions constant on depth- n dyadic cells. The Haar basis organizes this complement by location; the Walsh basis organizes it by sequency. $\square$

Thus the same rational E_n is simultaneously a local curvature statistic of F , a sibling-mass contrast, and a Walsh spectral-shell energy.

9.2 Local Taylor expansion of a Haar coefficient

Let $h = 2^{-n}$ and let $m_{n,k} = (k + 1/2)h$ be the center of a parent cell. Flatness of p at the endpoints permits uniform Taylor expansion to arbitrary order. Direct integration of odd Taylor terms gives

$$c_{n,k} \sim - \sum_{r=0}^{\infty} \frac{h^{2r+3/2}}{2^{2r+1}(2r+2)!} p^{(2r+1)}(m_{n,k}). \quad (180)$$

The first terms are

$$c_{n,k} = -h^{3/2} \left[\frac{p'(m_{n,k})}{4} + \frac{h^2 p'''(m_{n,k})}{192} + \frac{h^4 p^{(5)}(m_{n,k})}{23040} + \mathcal{O}(h^6) \right]. \quad (181)$$

This explains the scale $E_n \asymp 4^{-n}$ despite the infinite smoothness of p : the Haar wavelet has only one vanishing moment. Higher-order multiwavelets would expose the superalgebraic smoothness more efficiently.

9.3 A frequency multiplier and a superalgebraic alias remainder

Let

$$K_h(x) = h^{-1/2} (\mathbf{1}_{[-h/2,0]}(x) - \mathbf{1}_{[0,h/2]}(x)). \quad (182)$$

Its squared Fourier modulus is

$$|\widehat{K}_h(\xi)|^2 = \frac{16}{h\xi^2} \sin^4\left(\frac{h\xi}{4}\right). \quad (183)$$

The coefficients $c_{n,k}$ are samples of a translate of $p * K_h$ at the cell centers. Poisson summation for the squared samples separates a zero alias from all nonzero aliases.

Theorem 9.2 (sinc⁴ multiplier formula). *For $h = 2^{-n}$,*

$$E_n = \frac{1}{2\pi} \int_{\mathbb{R}} |\widehat{p}(\xi)|^2 \frac{h^2 \xi^2}{16} \text{sinc}^4\left(\frac{h\xi}{4}\right) d\xi + R(h), \quad (184)$$

where for every $M > 0$,

$$R(h) = \mathcal{O}_M(h^M) \quad (h \downarrow 0). \quad (185)$$

Proof. Apply Poisson summation to the samples of $|p * K_h|^2$ on the shifted lattice $h(\mathbb{Z} + 1/2)$. The zero Fourier mode is

$$\frac{1}{h} \|p * K_h\|_2^2 = \frac{1}{2\pi h} \int |\widehat{p}(\xi)|^2 |\widehat{K}_h(\xi)|^2 d\xi,$$

which simplifies to the integral in (184). Every nonzero alias contains a factor $\widehat{p}(\xi + 2\pi\ell/h)$ or its conjugate. Because the zero extension of p is $C_c^\infty(\mathbb{R})$, $\widehat{p}$ is Schwartz; the nonzero aliases are therefore smaller than every power of h after summation over $\ell \neq 0$. $\square$

9.4 All orders and Bell–Bernoulli coefficients

Write

$$\text{sinc}^4 z = \sum_{j=0}^{\infty} \alpha_j z^{2j}, \quad \alpha_0 = 1, \quad \alpha_1 = -\frac{2}{3}, \quad \alpha_2 = \frac{1}{5}, \quad \alpha_3 = -\frac{34}{945}, \dots \quad (186)$$

Expanding the multiplier in (184) and using Parseval gives:

Theorem 9.3 (All-orders Haar-energy expansion). *As $n \rightarrow \infty$, with $h = 2^{-n}$,*

$$E_n \sim \sum_{j=0}^{\infty} \frac{\alpha_j}{16^{j+1}} h^{2j+2} \|p^{(j+1)}\|_2^2. \quad (187)$$

In particular,

$$E_n \sim \frac{h^2}{16} \|p'\|_2^2 - \frac{h^4}{384} \|p''\|_2^2 + \frac{h^6}{20480} \|p'''\|_2^2 - \frac{17h^8}{30965760} \|p^{(4)}\|_2^2 + \dots. \quad (188)$$

Proof. Taylor-expand $\text{sinc}^4(h\xi/4)$ to an arbitrary fixed order in (184). The Schwartz decay of $\widehat{p}$ justifies termwise integration and bounds the remainder. Parseval identifies $(2\pi)^{-1} \int \xi^{2j+2} |\widehat{p}(\xi)|^2 d\xi$ with $\|p^{(j+1)}\|_2^2$. The alias term is smaller than every algebraic order. $\square$

The coefficients themselves are Bell–Bernoulli quantities. Since

$$\log \text{sinc}^4 z = - \sum_{r=1}^{\infty} \frac{2^{2r+1} |B_{2r}|}{r(2r)!} z^{2r}, \quad (189)$$

put

$$\beta_{2r} = - \frac{2^{2r+1} |B_{2r}|}{r}, \quad \beta_{2r+1} = 0. \quad (190)$$

Then

$$\alpha_j = \frac{1}{(2j)!} Y_{2j}(0, \beta_2, 0, \beta_4, \dots, \beta_{2j}). \quad (191)$$

Thus the same Bell–Bernoulli mechanism that governs the moment expansion of Φ also governs the continuum expansion of its dyadic Haar energies, but through the fourth power of a single-scale sinc filter.

9.5 Derivative norms from half-integer samples of the sinc product

The cosine series (39) gives exact rapidly convergent sums for every derivative norm:

$$\|p^{(r)}\|_2^2 = 2^{2r+1} \sum_{m=0}^{\infty} A_m^2 ((2m+1)\pi)^{2r}. \quad (192)$$

Substitution into (187) expresses the rational scale energies asymptotically through discrete samples of the infinite sinc product at the odd half-lattice. The first numerical values are reported in [section 13](#).

9.6 A meromorphic generating function for the rational energies

The complete scale expansion has a strong analytic consequence. Define

$$\mathcal{H}(z) = \sum_{n=0}^{\infty} E_n z^n. \quad (193)$$

Put

$$C_j = \frac{\alpha_j}{16^{j+1}} \|p^{(j+1)}\|_2^2. \quad (194)$$

Then (187) says, at every fixed depth J ,

$$E_n = \sum_{j=0}^J C_j 4^{-(j+1)n} + \mathcal{O}_J(4^{-(J+2)n}). \quad (195)$$

Theorem 9.4 (Meromorphic Haar-energy generating function). *The power series $\mathcal{H}$ has radius of convergence 4 and admits a meromorphic continuation to the whole complex plane. Its only possible finite poles are simple poles at*

$$z = 4, 16, 64, \dots, 4^{j+1}, \dots \quad (196)$$

Near $z = 4^{j+1}$ its principal part is

$$\frac{C_j}{1 - z/4^{j+1}}, \quad (197)$$

and hence the residue is

$$\text{Res}_{z=4^{j+1}} \mathcal{H}(z) = -\frac{\alpha_j}{4^{j+1}} \|p^{(j+1)}\|_2^2. \quad (198)$$

In particular,

$$\text{Res}_{z=4} \mathcal{H}(z) = -\frac{1}{4} \|p'\|_2^2. \quad (199)$$

Proof. The positive leading asymptotic $E_n \sim C_0/4^n$, with $C_0 = \|p'\|_2^2/16 > 0$, gives radius 4. For a fixed J , subtract from $\mathcal{H}$ the finite rational function

$$\sum_{j=0}^J \frac{C_j}{1 - z/4^{j+1}}.$$

By (195), the remaining coefficient sequence is $\mathcal{O}_J(4^{-(J+2)n})$, so its power series is holomorphic in $|z| < 4^{J+2}$. Letting J grow gives compatible continuations to every finite disc. The principal parts and residues are immediate. $\square$

This is a new $q = 1/4$ spectral object attached to the dyadic multiresolution, but unlike the zero determinant already present in the corpus its coefficients are rational sibling-mass energies. The pole at 4^{j+1} directly records the Sobolev norm of derivative order $j + 1$, multiplied by the Bell–Bernoulli coefficient α_j .

9.7 Beyond all orders: a phase-locked alias frontier

The remainder $R(2^{-n})$ is superalgebraic but not expected to be a featureless error term. Its Poisson representation samples cross-products of $\hat{p}$ at frequencies separated by $2\pi 2^n \ell$. The corpus proves that Φ has a log-square envelope with a nonconstant periodic phase in $\log_2 |t|$. Along a dyadic scale sequence, the phase of each fixed alias ℓ is locked at the fractional part of $\log_2(2\pi|\ell|)$.

Conjecture 9.5 (Phase-locked Haar alias transseries). *After subtraction of any fixed number of terms in (187), the first nonzero Poisson aliases admit a transseries whose exponential scale is determined by the proved real-axis decay of $\Phi(2\pi 2^n \ell)$ and whose amplitudes depend on the corresponding locked Lambert phase. In particular, the optimally truncated remainder should be asymptotic to a finite sum of $\ell = \pm 1$ alias contributions except at isolated cancellation phases.*

A proof would require inserting the phase-resolved Fourier asymptotic into the exact Poisson cross-correlation integral, not merely into a pointwise sample. This is a concrete meeting point between the repository’s Fourier-decay campaign and the new multiresolution energy theory.

10 Multiresolution of the inverse Fabius function

10.1 Faber–Schauder series and reflection

The inverse G is continuous and strictly increasing, so it has the universal Faber–Schauder expansion

$$G(u) = u + \sum_{n=0}^{\infty} \sum_{k=0}^{2^n-1} e_{n,k} \psi_{n,k}(u), \quad (200)$$

where

$$e_{n,k} = G\left(\frac{2k+1}{2^{n+1}}\right) - \frac{1}{2}G\left(\frac{k}{2^n}\right) - \frac{1}{2}G\left(\frac{k+1}{2^n}\right). \quad (201)$$

The sum is taken through complete dyadic levels and converges uniformly. From $G(1-u) = 1 - G(u)$,

$$e_{n,2^n-1-k} = -e_{n,k}. \quad (202)$$

Since F is convex on the left half and concave on the right half, G is concave on $[0, 1/2]$ and convex on $[1/2, 1]$; hence

$$e_{n,k} > 0 \quad (k < 2^{n-1}), \quad e_{n,k} < 0 \quad (k \geq 2^{n-1}). \quad (203)$$

The coefficients are generally not expected to be rational. They package the inverse-dyadic values studied elsewhere in the corpus into a global basis.

10.2 Fourier and Walsh images of the inverse

Transforming the tents gives, levelwise and locally uniformly,

$$\int_0^1 G(u) e^{-izu} du = \frac{e^{-iz}(1+iz) - 1}{z^2} + \sum_{n=0}^{\infty} \sum_{k=0}^{2^n-1} e_{n,k} 2^{-n-1} e^{-izm_{n,k}} \operatorname{sinc}^2\left(\frac{z}{2^{n+2}}\right). \quad (204)$$

This is the inverse analogue of (106). It is particularly well suited to low-frequency moments, because derivatives at $z = 0$ produce finite polynomial combinations of inverse-dyadic coefficients at each level.

The centered inverse $G(u) - 1/2$ is antisymmetric under $u \mapsto 1 - u$. Therefore its Walsh coefficients

$$g_m = \int_0^1 \left(G(u) - \frac{1}{2}\right) w_m(u) du \quad (205)$$

vanish for even Hamming weight. The inverse and the density thus have complementary Walsh parity:

$$a_m \neq 0 \Rightarrow r(m) \text{ even}, \quad g_m \neq 0 \Rightarrow r(m) \text{ odd}. \quad (206)$$

The finite-product transform (127) applies unchanged to the Walsh expansion of $G - 1/2$.

10.3 Boundary coefficients and slow variation

The leftmost coefficient is

$$e_{n,0} = G(2^{-n-1}) - \frac{1}{2}G(2^{-n}). \quad (207)$$

The phase-resolved Lambert endpoint theory in the repository implies the leading inverse scale

$$\log \frac{1}{G(y)} \sim \sqrt{2 \log 2 \log \frac{1}{y}} \quad (y \downarrow 0), \quad (208)$$

and, more strongly, the fixed-ratio slow-variation law

$$\frac{G(cy)}{G(y)} \rightarrow 1 \quad (y \downarrow 0), \quad c > 0. \quad (209)$$

Applying $c = 1/2$ to (207) gives a rigorous first-order consequence.

Proposition 10.1 (Boundary Schauder scale). *Assuming the proved phase-resolved inverse endpoint expansion from the primary corpus,*

$$e_{n,0} \sim \frac{1}{2}G(2^{-n}) = 2^{-\sqrt{2n}+o(\sqrt{n})}. \quad (210)$$

Thus the inverse Schauder coefficients at the boundary decay faster than every power of n but slower than every geometric sequence in the level n .

The crude leading model $G(2^{-n}) \approx 2^{-\sqrt{2n}}$ predicts one further term:

$$\frac{e_{n,0}}{G(2^{-n})} \approx \frac{1}{2} - \frac{\log 2}{\sqrt{2n}}. \quad (211)$$

The actual correction must include the lower-Lambert logarithmic terms and the nonconstant phase already known for the endpoint problem.

Conjecture 10.2 (Phase-resolved boundary coefficient). *There is an explicit expansion*

$$\frac{e_{n,0}}{G(2^{-n})} = \frac{1}{2} - \frac{\log 2}{\sqrt{2n}} + \frac{P_1(\log n, \Theta_n)}{n} + \frac{P_2(\log n, \Theta_n)}{n^{3/2}} + \dots, \quad (212)$$

where Θ_n is the exact lower-Lambert phase inherited from the endpoint expansion and each P_j is explicitly computable from its Bell-polynomial coefficient calculus.

10.4 Boundary domination of inverse interpolation

Let $L_N G$ denote the polygon through the inverse dyadic values $G(k2^{-N})$. Uniform continuity gives $L_N G \rightarrow G$, but the usual C^2 error bound fails because G' diverges at both endpoints. Concavity on the left shows that the interpolation error there is positive, and the first cell already has midpoint error $e_{N,0}$.

Conjecture 10.3 (Boundary-dominated inverse interpolation). *The global polygonal error is asymptotically attained in the two boundary cells:*

$$\|G - L_N G\|_\infty \sim e_{N,0} \sim \frac{1}{2} G(2^{-N}). \quad (213)$$

In particular,

$$\log_2 \|G - L_N G\|_\infty = -\sqrt{2N} + o(\sqrt{N}). \quad (214)$$

A proof should combine monotonicity of the inverse curvature on successive endpoint cells with the phase-resolved ratio asymptotics $G(2^{-N-j})/G(2^{-N})$. Even a two-sided constant-factor result would provide a sharp complexity theorem for inverse-Fabius interpolation.

11 Comparison of the representation families

The formulas developed above are not redundant. They expose different parts of the same function and have different numerical and formal strengths.

representation	finite data		building blocks	principal strength
Infinite convolution	dyadic scales	uniform	compact uniforms	Probability, support, smoothing equations, finite head–tail factorization.
Sinc product	no sampled values		$\text{sinc}(t/2^j)$	Zeros, entire-function structure, Fourier decay, canonical products.
Half-integer cosine	$\Phi((2m+1)\pi)$		cosines	Fast point evaluation, derivatives, Sobolev norms, twisted Poisson summation.
Dyadic cell mixture	rational $\Delta_{N,k}$		equal-width forms	uni- Positive local model, 4^{-N} transform quadrature, exact mass polynomial.

representation	finite data	building blocks	principal strength
Haar	rational $\gamma_{n,k}$	local sign steps	Curvature by scale and location, exact rational energies, meromorphic energy generating function.
Faber–Schauder	rational $d_{n,k}$	tents and sinc^2 atoms	Uniform piecewise-linear approximation of F and a Fourier series of translated sinc^2 functions.
Walsh–Hadamard	rational a_m	digital characters	Exact Hamming cancellation, Thue–Morse top mode, finite trigonometric-product transforms.
Bernstein/beta	rational $\Delta_{N,k}$	beta densities, ${}_1F_1$	Positive polynomial approximation and hypergeometric transform mixtures.
Quantile chords	inverse values $g_{N,k}$	variable-width uniforms	Equal probability cells, direct use of inverse Fabius values, finite sinc mixtures.
Legendre–Chebyshev	Bessel integrals of Φ	orthogonal polynomials	Global spectral approximation and transform-to-polynomial bridges.

11.1 A commuting diagram of finite transforms

At level N , one rational vector Δ_N enters three exact linear maps:

$$\Delta_N \xrightarrow{H_N} a^{(N)}, \quad \Delta_N \xrightarrow{\text{sibling differences}} \gamma_n, \quad \Delta_N \xrightarrow{\text{beta kernel}} p_N^B. \quad (215)$$

The same vector also determines the mass polynomial D_N . The resulting functions all integrate to one and converge to p , but their Fourier images factor differently:

$$\hat{p}_N(z) = \text{exponential polynomial} \times \text{sinc window}, \quad (216)$$

$$\hat{p}_N(z) = \sum_{m < 2^N} a_m \times \text{finite sine-cosine product}, \quad (217)$$

$$\hat{p}_N^B(z) = \sum_k \Delta_{N,k} \times {}_1F_1(k+1; 2^N+1; -iz). \quad (218)$$

This makes it possible to transfer a question among a local partition, a digital frequency basis, and a positive global polynomial basis without changing the exact arithmetic input.

11.2 Computational observations

For direct evaluation of $\Phi(t)$ at moderate t , the canonical sinc product remains the simplest representation. The new formulas are useful for other purposes:

- the cell and Haar formulas expose exact rationality and give certified dyadic approximation errors;
- the Walsh formula isolates cancellation order before numerical evaluation, avoiding subtraction of low moments;
- the beta mixture preserves positivity and is convenient for rigorous interval bounds on the real Laplace axis;
- the quantile chord formula is naturally adapted to a table of inverse values and does not require differentiating G ;

- the energy expansion converts fine-scale rational data into derivative norms and vice versa.

12 Further deductions and frontier directions

12.1 Exact arithmetic of multiresolution coefficients

The formulas

$$\begin{aligned}\gamma_{n,k} &= 2^n(\Delta_{n+1,2k} - \Delta_{n+1,2k+1}), \\ d_{n,k} &= \frac{1}{2}(\Delta_{n+1,2k} - \Delta_{n+1,2k+1}), \\ a_m &= \sum_k \varepsilon_{m,N}(k) \Delta_{N,k}\end{aligned}$$

reduce the arithmetic of three basis systems to exact dyadic Fabius values. The existing two-adic theory should therefore induce valuation laws for all three coefficient arrays.

Open problem 12.1 (Two-adic multiresolution spectrum). *Determine sharp formulas or lower envelopes for $v_2(\gamma_{n,k})$, $v_2(d_{n,k})$, and $v_2(a_m)$ in terms of binary carries, Hamming weight, and the known valuations of dyadic Fabius values. In particular, classify the indices for which sibling subtraction creates additional two-adic cancellation.*

A promising approach is to compute the exact mass vectors by the repository's binary reduction algorithm and apply the fast Walsh–Hadamard transform over $\mathbb{Q}_2$, retaining valuations before rational reconstruction.

12.2 Mass-polynomial renormalization

The palindromic rational polynomials D_N are projectively compatible, but (63) alone does not close them under a scalar renormalization. The functional equation of F supplies additional relations among sibling masses.

Open problem 12.2 (Closed recursion for D_N). *Find a matrix or q -difference recursion that computes D_{N+1} from a small number of transforms of D_N and finitely many boundary values, preferably in a form compatible with the existing q -binomial dyadic formulas. Such a recursion would turn (77) into an autonomous sequence of rational entire approximants.*

A two-component recursion separating the left and right halves is more plausible than a scalar recursion, because $F'(x) = 2F(2x)$ on the left couples a cell mass to an integral of F , not merely to another mass.

12.3 Hamming layers and Walsh pressure

Let

$$\mathcal{W}_{N,r} = \sum_{\substack{m < 2^N \\ r(m)=r}} a_m^2. \quad (219)$$

The exact parity filter removes odd r , and [theorem 6.6](#) controls the corresponding basis moments. The coefficient energy itself is a new digital spectrum.

Conjecture 12.3 (Walsh Hamming-layer pressure). *There exists a limiting pressure function $P(\theta)$ for the bivariate sums*

$$\sum_{m < 2^N} |a_m|^2 e^{\theta r(m)}, \quad (220)$$

possibly after subtraction of a deterministic quadratic scale term. Its Legendre transform should describe the distribution of Walsh energy across Hamming layers and should be related, but not identical, to the multifractal pressure already found for Fourier lobes.

The exact shell identity (179) supplies the marginal in binary length; the new problem is to resolve each shell by Hamming weight.

12.4 Higher-order rational multiwavelets

Haar has only one vanishing moment, explaining the geometric energy scale. Alpert-type multiwavelets on dyadic cells have any prescribed number r of vanishing moments. Their coefficients are finite linear combinations of integrals $\int_I x^j p(x) dx$, which the repository's antiderivative and exact dyadic machinery can evaluate rationally.

Open problem 12.4 (Fabius–Alpert hierarchy). *Construct an exact rational multiwavelet hierarchy for p and derive its Fourier product-series. For order r , the detail energy should start at $h^{2r} \|p^{(r)}\|_2^2$ with a multiplier determined by the corresponding finite filter. Determine the Bell–Bernoulli coefficients and the meromorphic pole lattice of its scale-energy generating function.*

This would replace the single sinc^4 multiplier by squared higher-order finite-difference symbols and produce a family of rational spectral objects.

12.5 Beta mixtures and confluent dyadic extrapolation

The beta mixture converges at the binomial scale $n^{-1} = 2^{-N}$, slower than the cell transform's 4^{-N} rate. Its complete Voronovskaya expansion is in integer powers of 2^{-N} , so the repository's confluent geometric filters can be applied without abandoning positivity at the base level.

Open problem 12.5 (Accelerated hypergeometric mixtures). *Derive the full operator expansion of $\sum_k \Delta_{N,k} F_1(k+1; 2^N+1; z)$ on compact z -sets and construct $q = 1/2$ extrapolation filters that cancel the first r Bernstein corrections. Quantify how acceleration affects positivity, conditioning, and zero locations.*

12.6 Quantile-width extremes and Lambert phases

The largest quantile width is the boundary width $q_{N,0} = G(2^{-N})$, up to symmetry. Hence the extremes of the projective array q_N are governed by the same Lambert phase as the endpoint inverse. The dual Rényi spectrum diverges for $s > 1$ because these boundary widths are much larger than 2^{-N} .

Conjecture 12.6 (Extreme-width partition function). *For $s > 1$,*

$$\sum_{k=0}^{2^N-1} q_{N,k}^s \sim 2G(2^{-N})^s \mathcal{C}_s(\Theta_N), \quad (221)$$

where $\mathcal{C}_s$ is a positive phase-dependent correction obtained by summing a finite boundary layer $G((j+1)2^{-N}) - G(j2^{-N})$ and Θ_N is the lower-Lambert phase.

This would turn inverse endpoint asymptotics into a discrete partition function and provide a phase-resolved refinement of the L^s threshold for G' .

12.7 Base- b and nonuniform scale systems

The multiresolution arguments do not fundamentally require base two. A base- b weighted-uniform series would produce b -adic mass arrays, b -ary Haar systems, and characters of $(\mathbb{Z}/b\mathbb{Z})^N$. The Walsh moment cancellation would be replaced by digit-character cancellation, while q -binomial aggregates would naturally involve $q = 1/b$.

Open problem 12.7 (Base- b cancellation spectrum). *For the geometric scale b^{-j} , derive the analogue of theorems 6.5 and 6.6, identify the first nonzero moments of b -ary characters, and determine which Gaussian or multinomial q -coefficients aggregate the digit-position factors.*

For nongeometric weights the finite character product still exists, but the clean q -binomial aggregation is replaced by an elementary symmetric function of the scales.

12.8 Formalization route

Most new statements separate into finite algebra plus standard convergence:

1. define dyadic mass vectors and prove projective consistency;
2. formalize the finite Hadamard matrix and the identity $H_N H_N^T = 2^N I$;
3. identify Walsh partial sums with dyadic conditional expectations;
4. prove the finite product (126) by induction on bit length;
5. extract vanishing moments from the order of zero at the origin;
6. use the finite q -binomial theorem for (136);
7. obtain beta mixtures by differentiating the already elementary Bernstein polynomial;
8. formalize quantile chord integration as an affine integral on each cell.

The analytic bottleneck is the all-orders alias estimate, which requires a Schwartz-space Poisson-summation API. The finite identities and all rationality claims should be substantially easier to integrate into Lean.

13 Numerical and exact-symbolic verification

The accompanying script `numerical_experiments.py` is not used as a substitute for proof. It serves four narrower purposes:

1. independently check the Fourier conventions and finite-product signs;
2. verify the exact Walsh moment and q -binomial identities over a large finite range using rational arithmetic;
3. compare four finite transform representations against the canonical sinc product;
4. test the constants in the Haar-energy expansion.

The default run uses 55 decimal digits, 160 sinc factors, and 200 half-integer cosine modes. It completes in a few seconds in the supplied environment and writes both a plain-text ledger and two figures.

13.1 Exact checks

SymPy rational arithmetic checked all vanishing moments and first nonzero moments for every nonzero Walsh index of binary length at most seven:

$$1016 \text{ exact moment identities checked, } \quad 0 \text{ failures.} \quad (222)$$

It also checked all 35 pairs (N, r) with $1 \leq N \leq 7$ in (136), again with zero failures. Selected exact moments are

m	binary m	$r(m)$	$\int_0^1 x^{r(m)} w_m(x) dx$
1	1	1	$-1/4$
2	10	1	$-1/8$
3	11	2	$1/16$
5	101	2	$1/32$
6	110	2	$1/64$
7	111	3	$-3/256$
15	1111	4	$3/2048$
31	11111	5	$-15/131072$

At the complex test point $z = 3.7 + 0.4i$, the largest discrepancy between the cell-by-cell integral of W_m and the finite product (127), for $0 \leq m \leq 15$, was 3.75×10^{-56} . At dyadic level seven, the largest numerically computed coefficient with odd Hamming weight was 1.73×10^{-54} , while finite Hadamard Parseval had zero residual at the working precision.

As a normalization check, the cosine evaluator gives

$$F(1/4) = 0.0694444444444445713\dots, \quad \frac{5}{72} = 0.06944444444444444\dots;$$

the 1.27×10^{-16} difference is the deliberately finite half-integer Fourier tail, not a failure of the exact dyadic value.

13.2 Transform convergence

The test frequencies were $t = 0.7, 3.1, 7.3, 11.2$. The table reports the maximum absolute error relative to $\Phi(t) = \prod_{j \geq 1} \text{sinc}(t/2^j)$.

N	cell/Walsh	midpoint atomic	beta mixture	Walsh product series
2	$9.720 \cdot 10^{-2}$	$8.750 \cdot 10^{-1}$	$3.312 \cdot 10^{-1}$	$9.720 \cdot 10^{-2}$
3	$2.780 \cdot 10^{-2}$	$1.416 \cdot 10^{-2}$	$1.985 \cdot 10^{-1}$	$2.780 \cdot 10^{-2}$
4	$7.012 \cdot 10^{-3}$	$3.517 \cdot 10^{-3}$	$1.098 \cdot 10^{-1}$	$7.012 \cdot 10^{-3}$
5	$1.760 \cdot 10^{-3}$	$8.805 \cdot 10^{-4}$	$5.796 \cdot 10^{-2}$	$1.760 \cdot 10^{-3}$
6	$4.403 \cdot 10^{-4}$	$2.202 \cdot 10^{-4}$	$2.980 \cdot 10^{-2}$	$4.403 \cdot 10^{-4}$
7	$1.101 \cdot 10^{-4}$	$5.505 \cdot 10^{-5}$	$1.512 \cdot 10^{-2}$	$1.101 \cdot 10^{-4}$

The equality of the cell and Walsh errors is exact, not accidental: the power-of-two Walsh partial sum is precisely p_N , whose transform is the cell mixture. The observed factor near four per level agrees with (81). The Bernstein/beta approximation displays its slower factor near two per level.

13.3 Haar energy asymptotics

The half-integer Fourier samples give

$$\|p'\|_2^2 = 25.8839531509750580138\dots, \quad (223)$$

$$\|p''\|_2^2 = 1656.57300166240371284\dots, \quad (224)$$

$$\|p'''\|_2^2 = 424082.688425575063737\dots \quad (225)$$

The exact rational energies and the three-term approximation from (188) behave as follows.

n	E_n	$16 \cdot 4^n E_n$	relative error of three terms
2	$6.2500000 \cdot 10^{-2}$	16.0000000000	$4.290 \cdot 10^{-1}$
3	$2.5292927 \cdot 10^{-2}$	25.8999572626	$3.914 \cdot 10^{-2}$
4	$6.2471084 \cdot 10^{-3}$	25.5881558642	$1.220 \cdot 10^{-3}$
5	$1.5757619 \cdot 10^{-3}$	25.8172829143	$1.627 \cdot 10^{-5}$
6	$3.9470093 \cdot 10^{-4}$	25.8671202723	$4.231 \cdot 10^{-8}$
7	$9.8723379 \cdot 10^{-5}$	25.8797415033	$4.139 \cdot 10^{-11}$

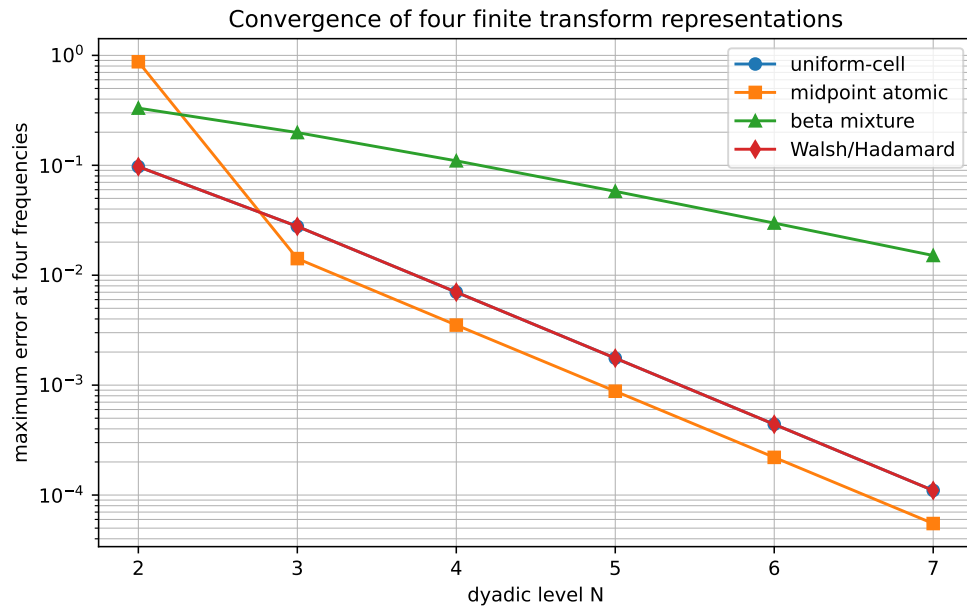


Figure 1: Convergence of the fixed-cell, midpoint-atomic, beta-mixture, and Walsh–Hadamard transform representations at four sample frequencies.

The scaled energy converges to $\|p'\|_2^2$, while the three-term expansion quickly becomes much more accurate than the displayed leading convergence.

13.4 Reproducibility files

The archive contains:

file	purpose
numerical_experiments.py	fully commented exact and numerical checks
numerical_results.txt	complete deterministic output ledger
multiresolution_convergence.pdf/.png	transform convergence figure
haar_energy_asymptotics.pdf/.png	scale-energy figure

The script contains independent direct and product implementations of the Walsh transform, rather than comparing a formula with a rearrangement of itself.

14 Conclusions

The central organizing object of this report is the exact dyadic mass array $\Delta_{N,k}$. It simultaneously yields:

1. a positive uniform-cell approximation and a rational exponential-polynomial transform limit;
2. rational unnormalized Haar coefficients and rational Faber–Schauder coefficients;
3. exact Walsh–Hadamard coefficients, with a finite sine–cosine product for every basis transform;
4. a positive beta-density mixture and a confluent-hypergeometric transform limit;
5. exact rational scale energies whose generating function has poles at the geometric lattice $4, 16, 64, \dots$

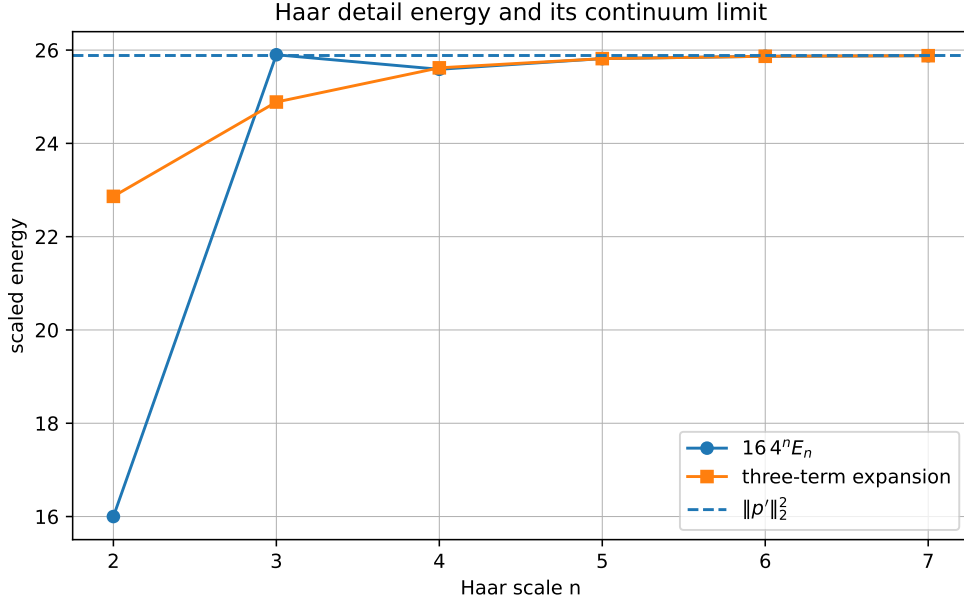


Figure 2: The scaled exact Haar energy, its three-term continuum expansion, and the limiting derivative norm.

The dual inverse array $q_{N,k}$ exchanges fixed widths and variable weights for fixed weights and variable widths. Its chord integrals give finite sinc mixtures for Φ and complete homogeneous polynomial limits for the Fabius moments.

The strongest new structural bridge is

binary Hamming weight $\longleftrightarrow$ Walsh vanishing moments

$$\longleftrightarrow \left[\begin{matrix} N \\ r \end{matrix} \right]_{1/2} \longleftrightarrow \text{Bell–Bernoulli higher moments.}$$

It complements the repository’s existing $q = 1/4$ moment and scale structures. The strongest analytic bridge is

rational Haar detail energies $\longleftrightarrow \text{sinc}^4$ multiplier

$$\longleftrightarrow \|p^{(r)}\|_2^2 \longleftrightarrow \text{meromorphic poles at } 4^r.$$

Both bridges are finite and exact before limits are taken, making them good candidates for formalization.

The most promising next steps are the two-adic classification of the coefficient arrays, a higher-order rational multiwavelet hierarchy, phase-resolved asymptotics of the inverse boundary coefficients, and an exact analysis of the first Poisson aliases beyond the sinc^4 energy expansion.

A Compact formula index

object	equation	representation type
up	(12)	infinite convolution of dyadic uniforms
up	(13)	infinite-dimensional delta integral
up	(15)	limit of finite Thue–Morse splines
up	(19)	finite spline convolved with a scaled copy

object	equation	representation type
Φ	(20)	infinite sinc product
Φ	(24)	q -Pochhammer product at $q = 1/4$
Φ	(25)	canonical zero product with $1 + v_2(n)$ multiplicity
$\log \Phi$	(26)	zeta power series
up	(34)	Fourier cosine integral
F	(35)	sine-over-frequency integral
F	(40)	half-integer cosine series
p	(41)	half-integer sine series
$\mathcal{F}_F$	(43)	compact transform through Φ
Φ	(47)	inverse-Fabius quantile integral
up	(55)	Fourier–Legendre/Bessel coefficient integral
up	(57)	Chebyshev/Bessel coefficient integral
p	(90)	rational unnormalized Haar series
F	(100)	rational Faber–Schauder series
$\mathcal{F}_F$	(106)	translated sinc ² product-series
p	(116)	rational Walsh–Hadamard series
Φ	(129)	series of finite sine–cosine products
Walsh moments	(144)	Bell–Bernoulli finite-product coefficients
F	(149)	limit of rational Bernstein polynomials
p	(153)	positive rational beta mixture
Φ	(156)	confluent-hypergeometric mixture limit
Φ	(77)	fixed-width rational sinc mixture
Φ	(164)	equal-weight variable-width sinc mixture
Fabius moments	(85)	rational fixed-cell polynomial limit
Fabius moments	(158)	rational beta-moment limit
Fabius moments	(169)	inverse-dyadic polygon limit
E_n	(184)	sinc ⁴ Fourier multiplier plus aliases
E_n	(187)	Bell–Bernoulli Sobolev expansion
$\mathcal{H}$	(193)	meromorphic rational-energy generating series
G	(200)	inverse Faber–Schauder series
$\hat{G}$	(204)	inverse translated-sinc ² series

B Proof details for two auxiliary identities

B.1 Derivation of the local Haar expansion

Let m be the center of a parent interval of length h . Then

$$\begin{aligned}
c_{n,k} &= h^{-1/2} \left(\int_{m-h/2}^m p(x) \, dx - \int_m^{m+h/2} p(x) \, dx \right) \\
&= -h^{-1/2} \sum_{r \geq 0} \frac{2p^{(2r+1)}(m)}{(2r+1)!} \frac{(h/2)^{2r+2}}{2r+2}
\end{aligned}$$

$$= - \sum_{r \geq 0} \frac{h^{2r+3/2}}{2^{2r+1}(2r+2)!} p^{(2r+1)}(m),$$

which is (180). Squaring and summing midpoint samples recovers the first coefficients in (188); the multiplier proof packages all integrations by parts into one sinc^4 symbol.

B.2 Absolute convergence of the first-moment Hamming layers

Letting $N \rightarrow \infty$ in (136) gives

$$\sum_{\substack{m \geq 0 \\ r(m)=r}} 2^{-\sigma(m)} = \frac{2^{-r(r-1)/2}}{(1/2; 1/2)_r}. \quad (226)$$

Since $|a_m| \leq \int_0^1 p = 1$, the weight- r contribution formed from the first nonzero Walsh moments is absolutely summable. In particular, the sum rule (146) is independent of the blockwise ordering.

C Reproducibility ledger

Dyadic Multiresolution and Product-Series Representations				
Numerical and exact-symbolic verification				
=====				
mpmath decimal digits	:	55		
sinc-product factors	:	160		
half-integer Fourier modes	:	200		
1. Normalizations				

F(1/4)	=	0.06944444444444445713227823382417		
Expected F(1/4)	=	0.0694444444444444444444444444444444		
absolute residual	=	1.2687834e-16		
sum of level-7 masses	=	1.0		
2. Exact Walsh moment and q-binomial checks				

Walsh moment identities	:	1016 checked, 0 failed		
q-binomial aggregates	:	35 checked, 0 failed		
Selected first nonzero Walsh moments:				
m	binary	r	integral $x^r w_m(x) dx$	
1	1	1	-1/4	
2	10	1	-1/8	
3	11	2	1/16	
5	101	2	1/32	
6	110	2	1/64	
7	111	3	-3/256	
15	1111	4	3/2048	
23	10111	4	3/4096	
31	11111	5	-15/131072	
3. Walsh finite-product transform				

max residual $m=0, \dots, 15$ at $z=3.7+0.4i$:		3.745842e-56		
4. Symmetry selection rule for Walsh coefficients				

max $ a_m $, odd Hamming weight (should be 0):		1.720445e-54		
max $ a_m $, nonzero even weight	:	0.7222222222222221715		
Hadamard Parseval residual, level 7	:	0.0		
5. Transform convergence				

level	max uniform-cell error	max atomic error	max beta-mixture error	max Walsh error
2	0.09719740288	0.8749804357	0.3311617664	0.09719740288
3	0.0278015789	0.01416477865	0.1984606384	0.0278015789
4	0.007011640078	0.003516689896	0.1097810782	0.007011640078

5	0.001759556884	0.0008804669754	0.05795858183	0.001759556884
6	0.0004403020799	0.0002201940863	0.02980477899	0.0004403020799
7	0.0001101013485	5.505336511e-5	0.01511621869	0.0001101013485

6. Haar energy and all-orders continuum expansion

$ p' _2^2$	=	25.8839531509750580137998	
$ p'' _2^2$	=	1656.57300166240371284162	
$ p''' _2^2$	=	424082.688425575063736863	
n	E_n	16*4^n E_n	rel.error(3-term)
2	0.0625	16.000000000000006	0.42900991
3	0.025292927014271	25.899957262613	0.039135771
4	0.0062471083652827	25.58815586419799	0.0012204175
5	0.0015757618966264	25.81728291432679	1.6265272e-5
6	0.00039470093189003	25.86712027234482	4.2313174e-8
7	9.872337914768e-5	25.87974150328945	4.1389921e-11

7. Low-level rational-structure samples (high-precision decimals)

level 1: Haar $c_{\{n,k\}}$ = -0.510688230857, 0.510688230857
Schauder $d_{\{n,k\}}$ = -0.180555555556, 0.180555555556

level 2: Haar $c_{\{n,k\}}$ = -0.125, -0.125, 0.125, 0.125
Schauder $d_{\{n,k\}}$ = -0.03125, -0.03125, 0.03125, 0.03125

level 3: Haar $c_{\{n,k\}}$ = -0.00943081845124, -0.0789575291971, -0.0789575291971, -0.00943081845124,
0.00943081845124, 0.0789575291971, 0.0789575291971, 0.00943081845124
Schauder $d_{\{n,k\}}$ = -0.00166714891975, -0.0139578510802, -0.0139578510802, -0.00166714891975,
0.00166714891975, 0.0139578510802, 0.0139578510802, 0.00166714891975

level 4: Haar $c_{\{n,k\}}$ = -0.000271267361112, -0.00808376736111, -0.0231662326389, -0.0309787326389,
-0.0309787326389, -0.0231662326389, -0.00808376736111, -0.00027126736111, 0.00027126736111,
0.00808376736111, 0.0231662326389, 0.0309787326389, 0.0309787326389, 0.0231662326389,
0.00808376736111, 0.000271267361112
Schauder $d_{\{n,k\}}$ = -3.3908420139e-5, -0.00101047092014, -0.00289577907986,
-0.00387234157986, -0.00387234157986, -0.00289577907986, -0.00101047092014, -3.39084201388e-5,
3.39084201388e-5, 0.00101047092014, 0.00289577907986, 0.00387234157986, 0.00387234157986,
0.00289577907986, 0.00101047092014, 3.3908420139e-5

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